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Evans et al.

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(54) **THERMAL OXIDIZATION SYSTEMS AND METHODS WITH GREENHOUSE GAS CAPTURE**

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B01D 53/44 (2006.01)

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(52) **U.S. Cl.**

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CPC F23G 7/063; F23G 2202/101; F23G 2207/101; F23D 14/62; F23N 5/102; B01D 2257/708

See application file for complete search history.

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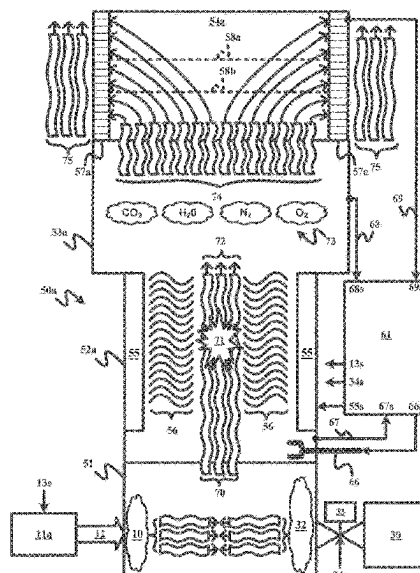
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(57) **ABSTRACT**

A thermal oxidizer employing an oxidation mixer, an oxidation chamber, a retention chamber and a heat dissipater forming a fluid flow path for thermal oxidation of a waste gas. In operation, the oxidation mixer facilitates a combustible mixture of the waste gas and an oxidant into an combustible waste gas stream, the oxidation chamber facilitates a primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream, the retention chamber facilitates a secondary combustion reaction of the oxygenated waste gas stream into oxidized gases and the heat dissipater reduces the temperature of the flow of oxidized gases within the heat dissipator, which is communicated to a greenhouse gas processor that extracts greenhouse gas(es) from the vaporized oxidized gases. The greenhouse gas processor may condensate the greenhouse gas(es), acid neutralize the condensation of the greenhouse gas(es), and capture the acid neutralization of the condensation of the greenhouse gas(es).

20 Claims, 26 Drawing Sheets



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F23N 1/02 (2006.01)

(52) **U.S. Cl.**

CPC *B01D 2258/02* (2013.01); *F23G 2207/101* (2013.01); *F23G 2207/20* (2013.01); *F23G 2207/30* (2013.01)

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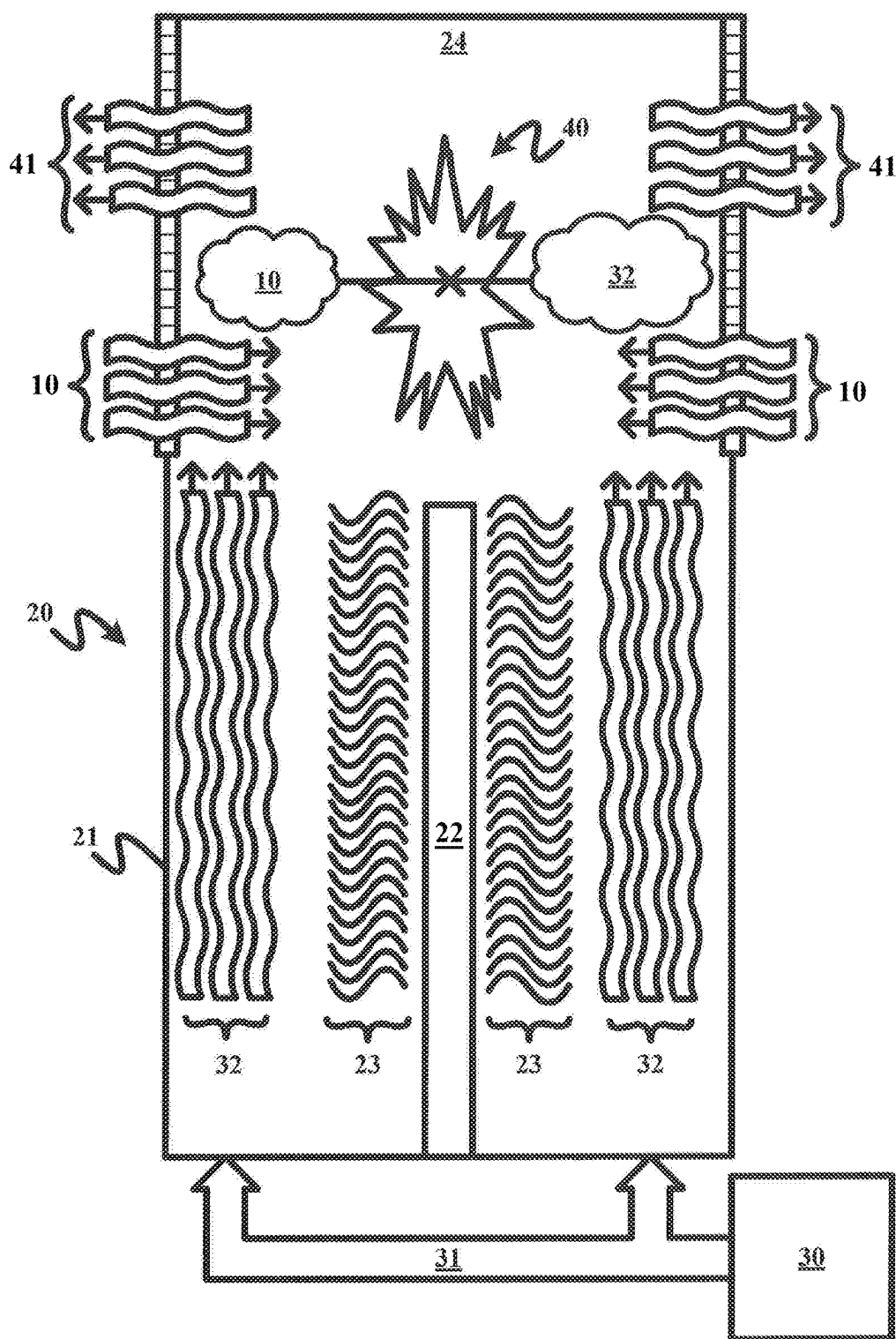


FIG. 1
(PRIOR ART)

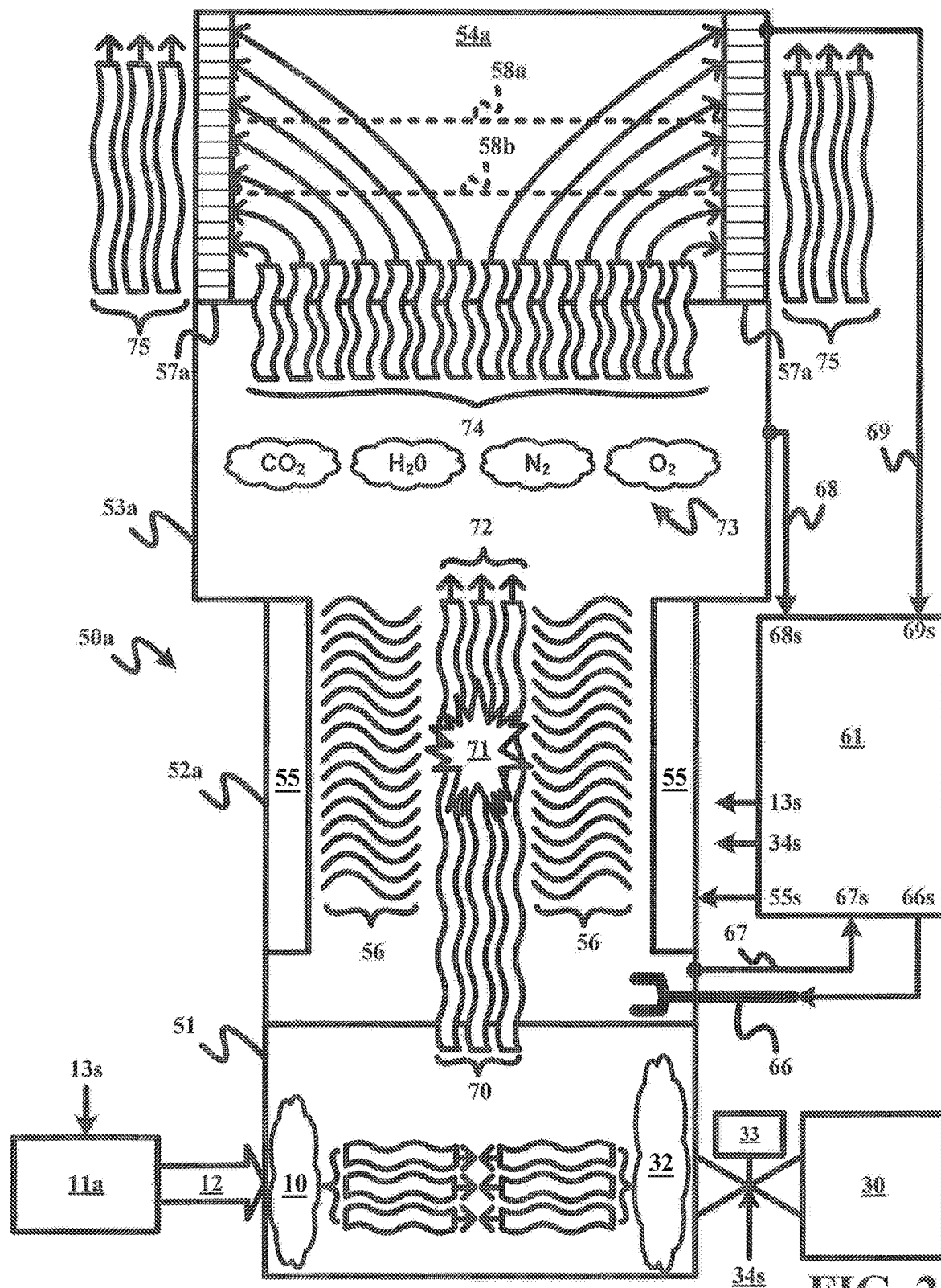


FIG. 2

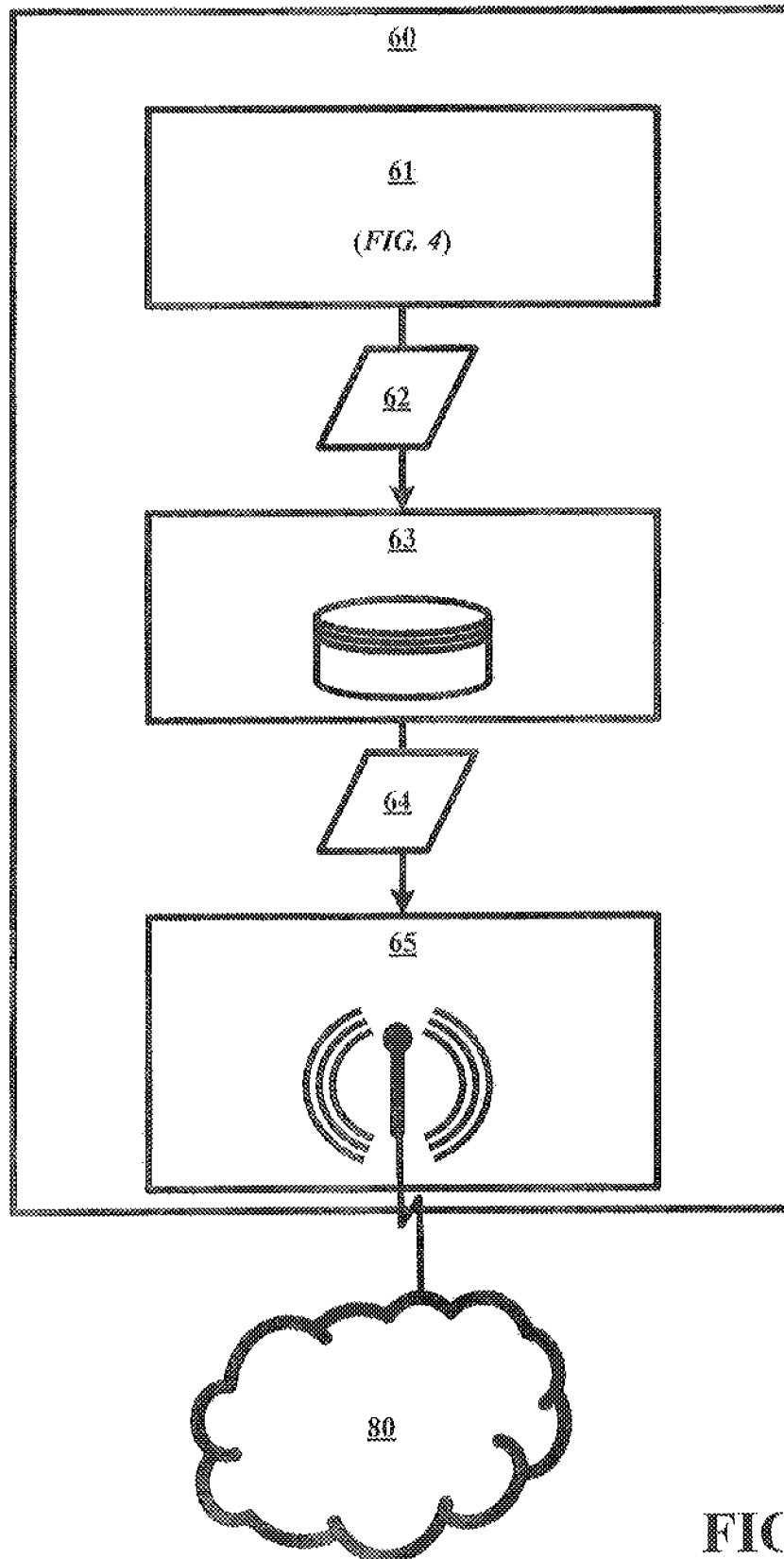
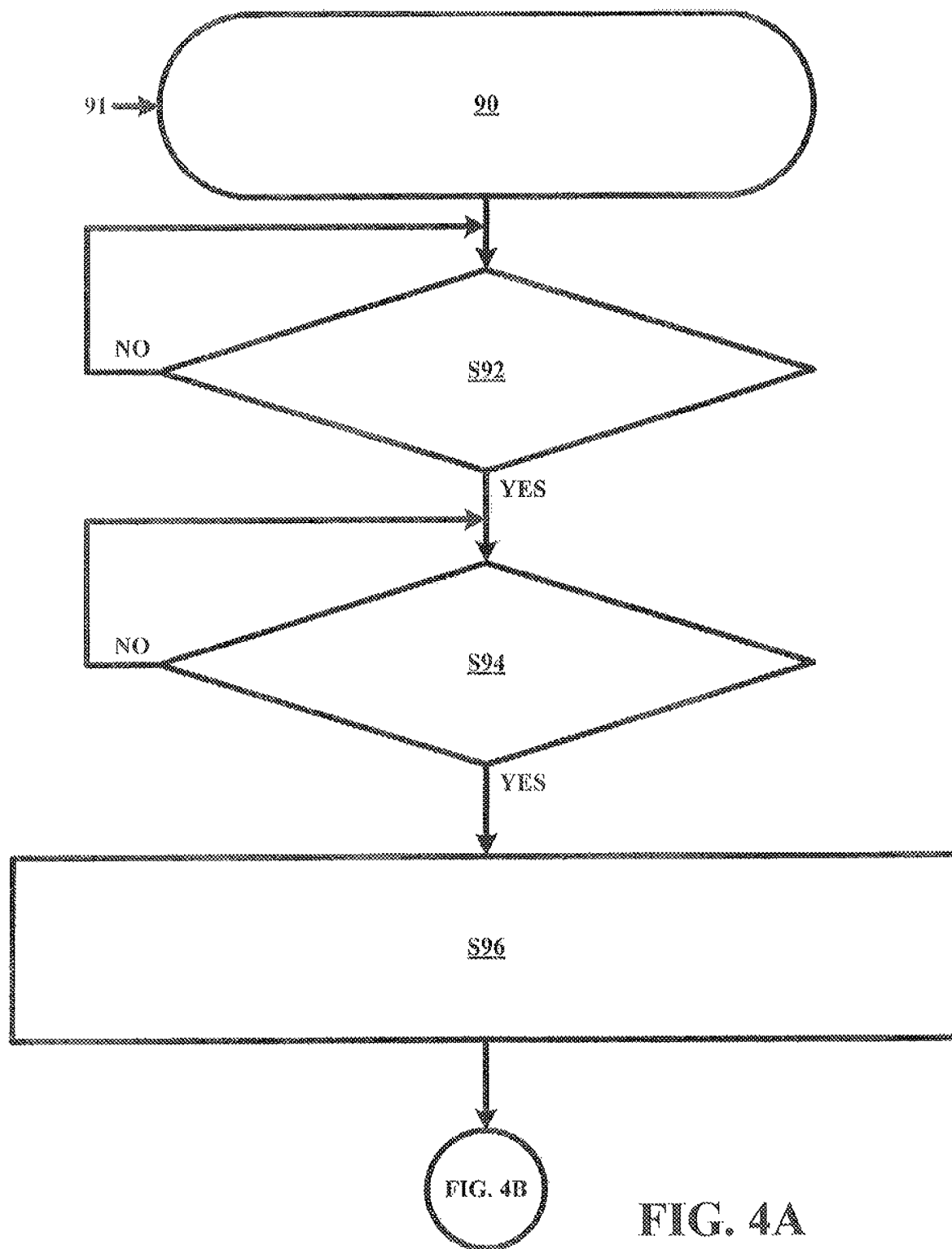


FIG. 3



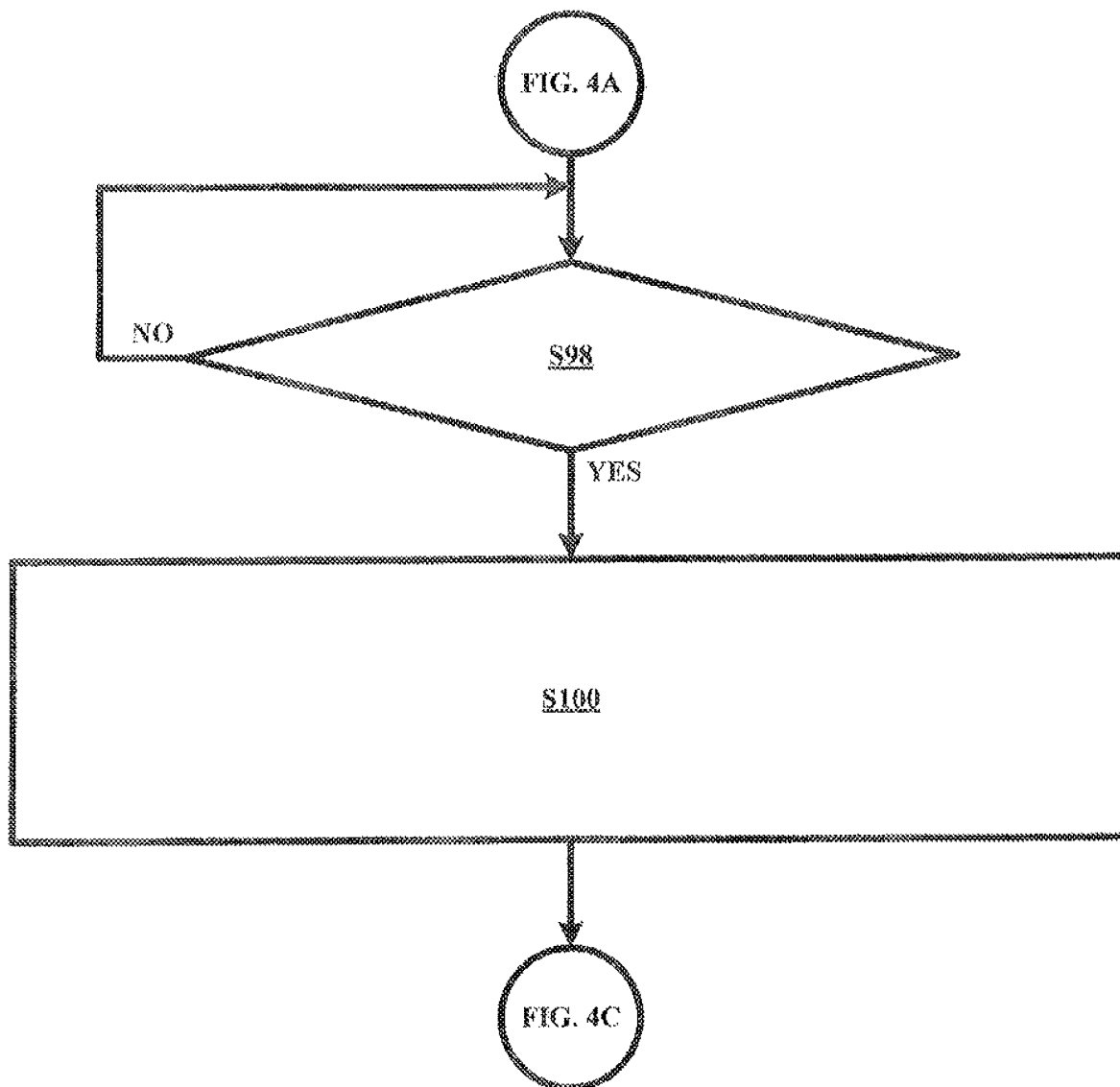


FIG. 4B

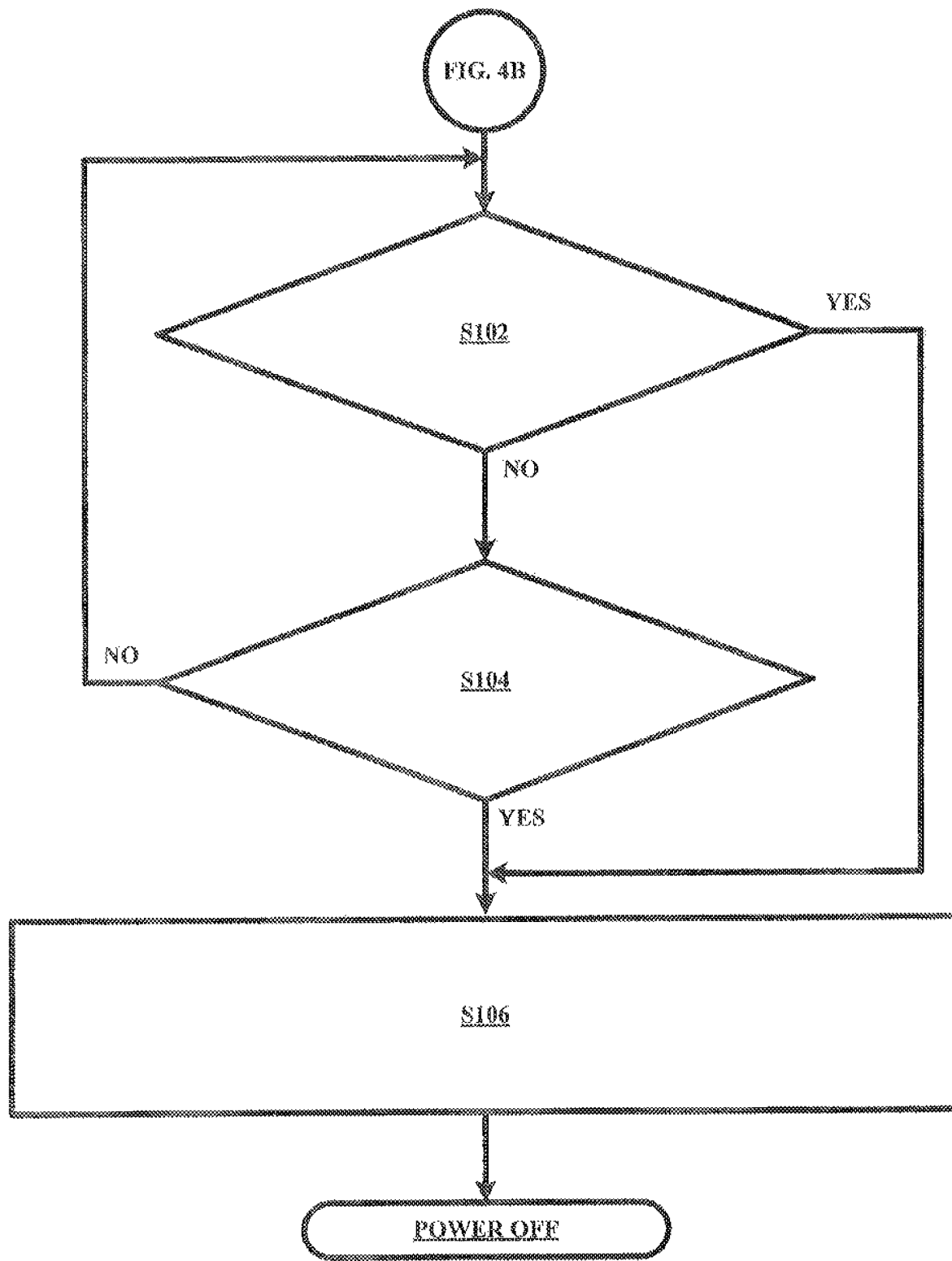


FIG. 4C

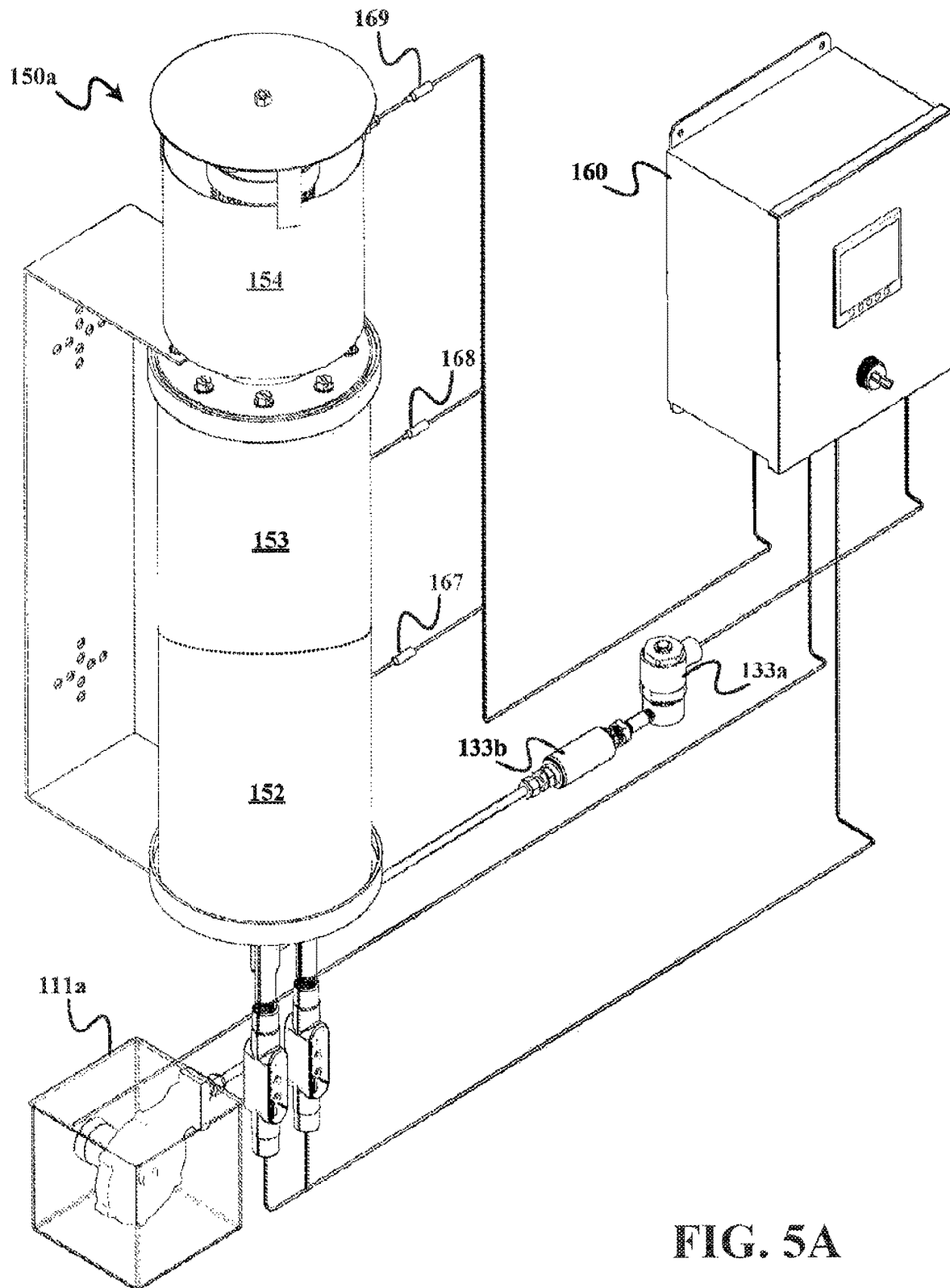
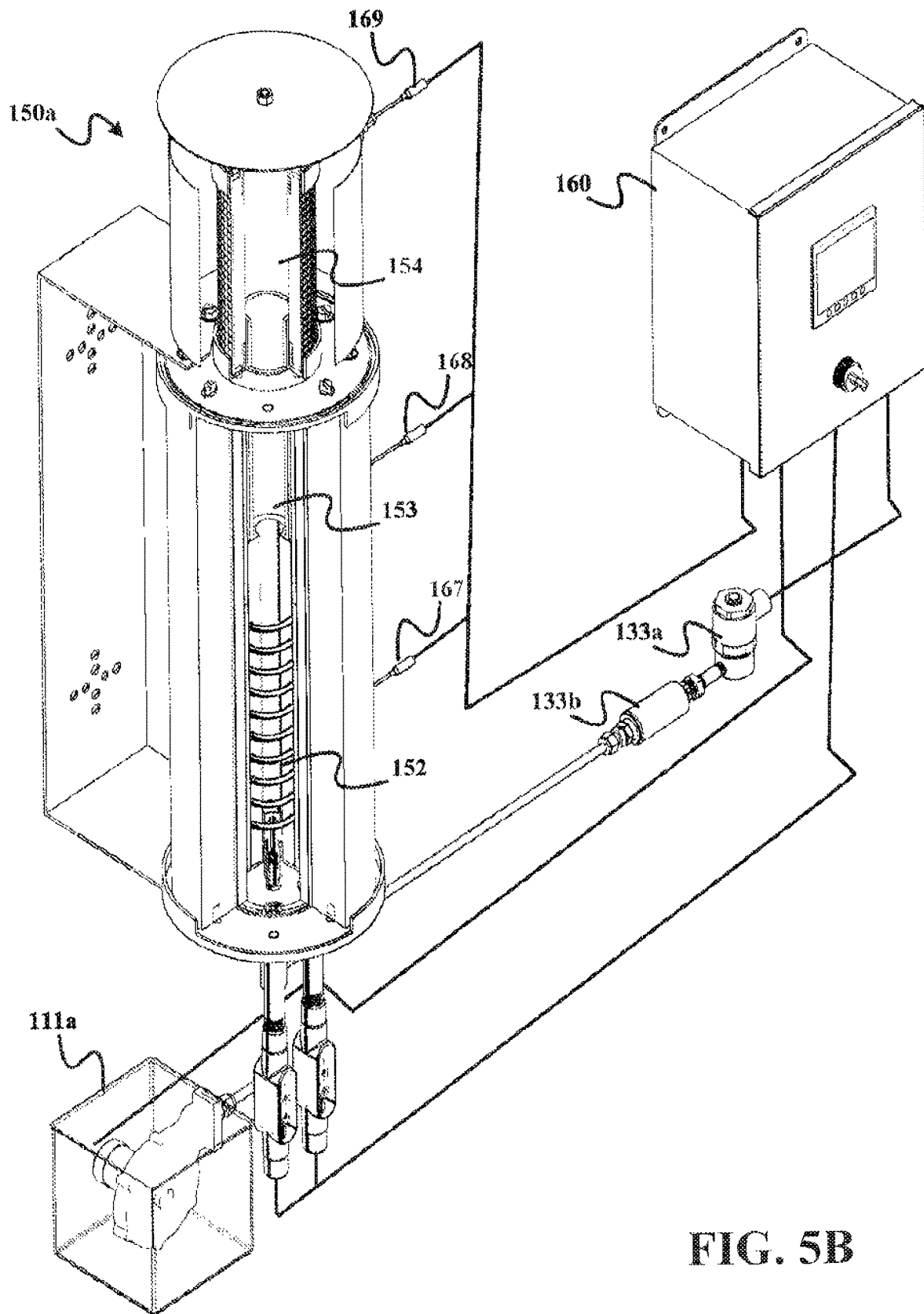
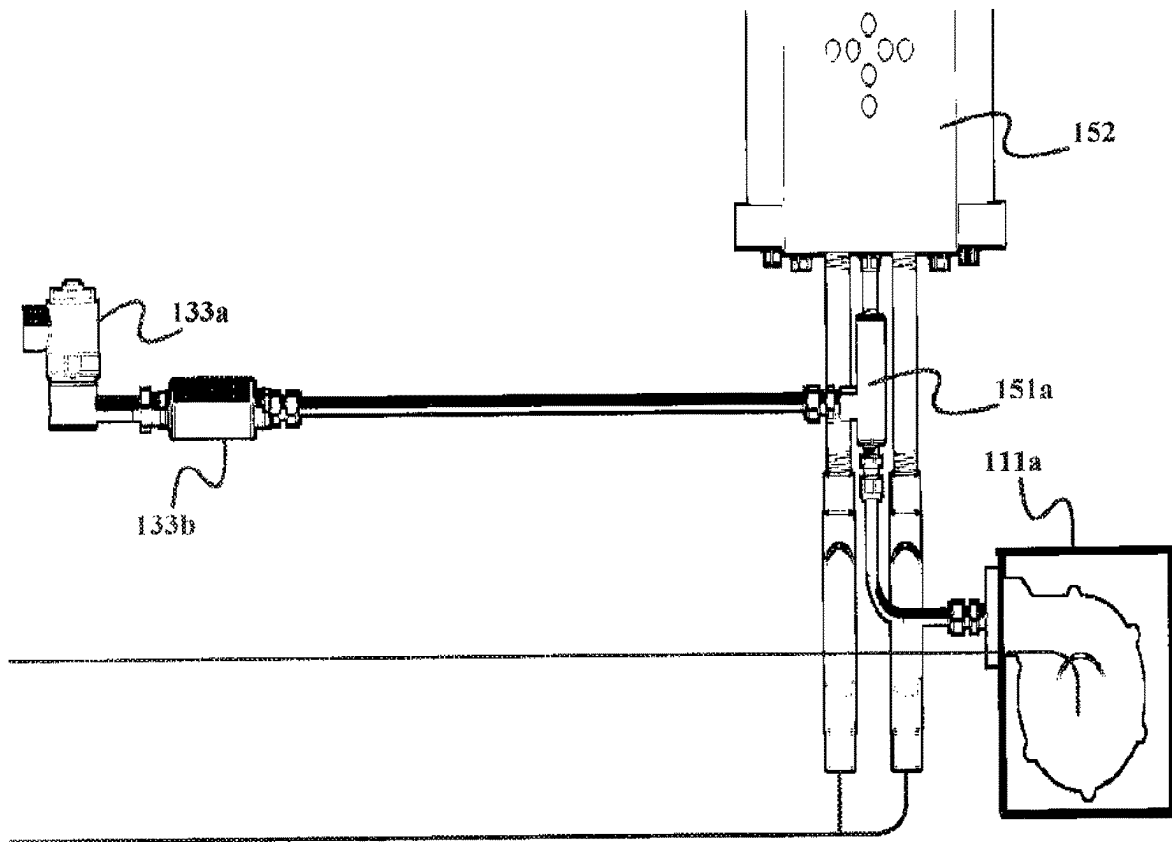
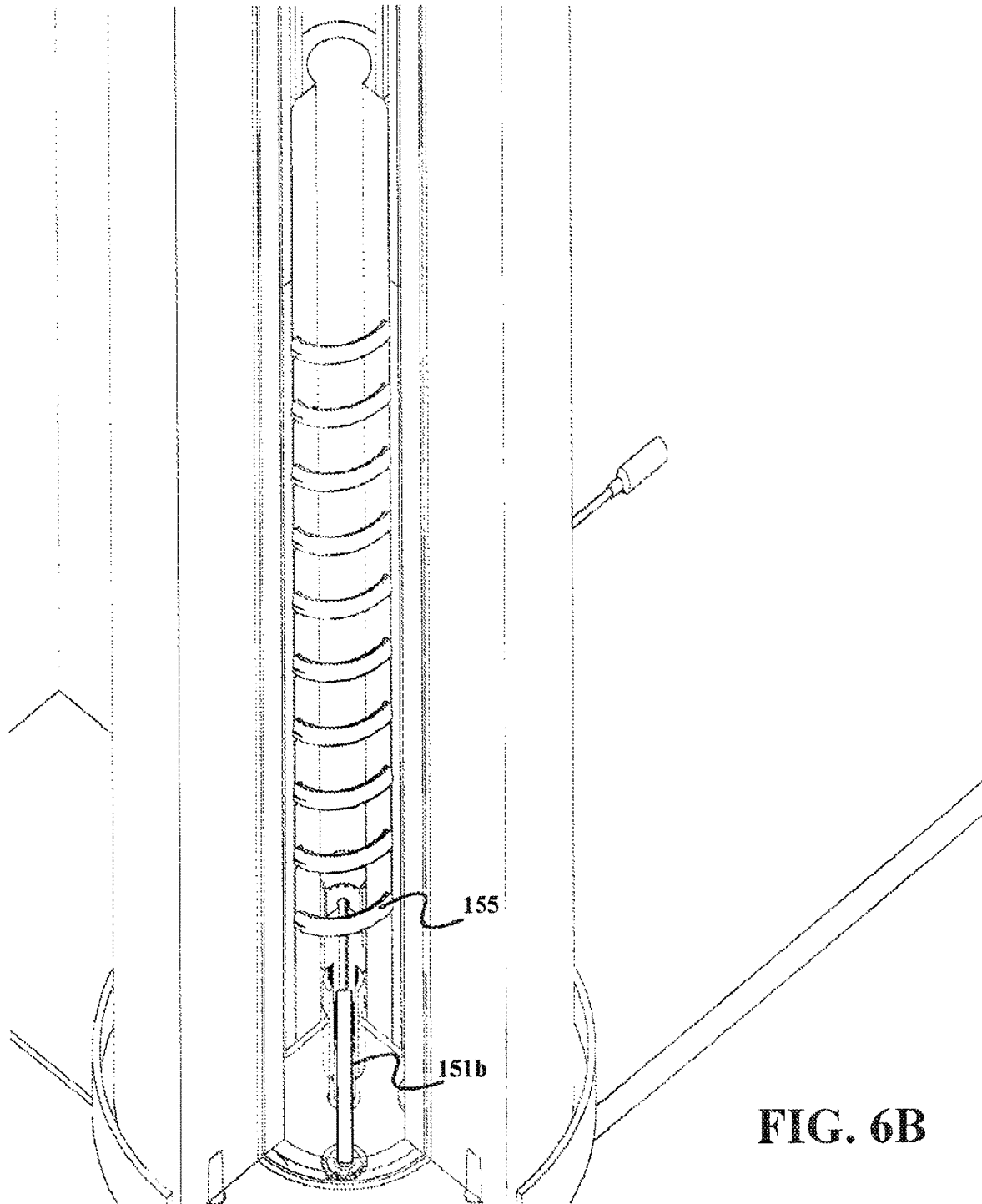


FIG. 5A



**FIG. 6A**



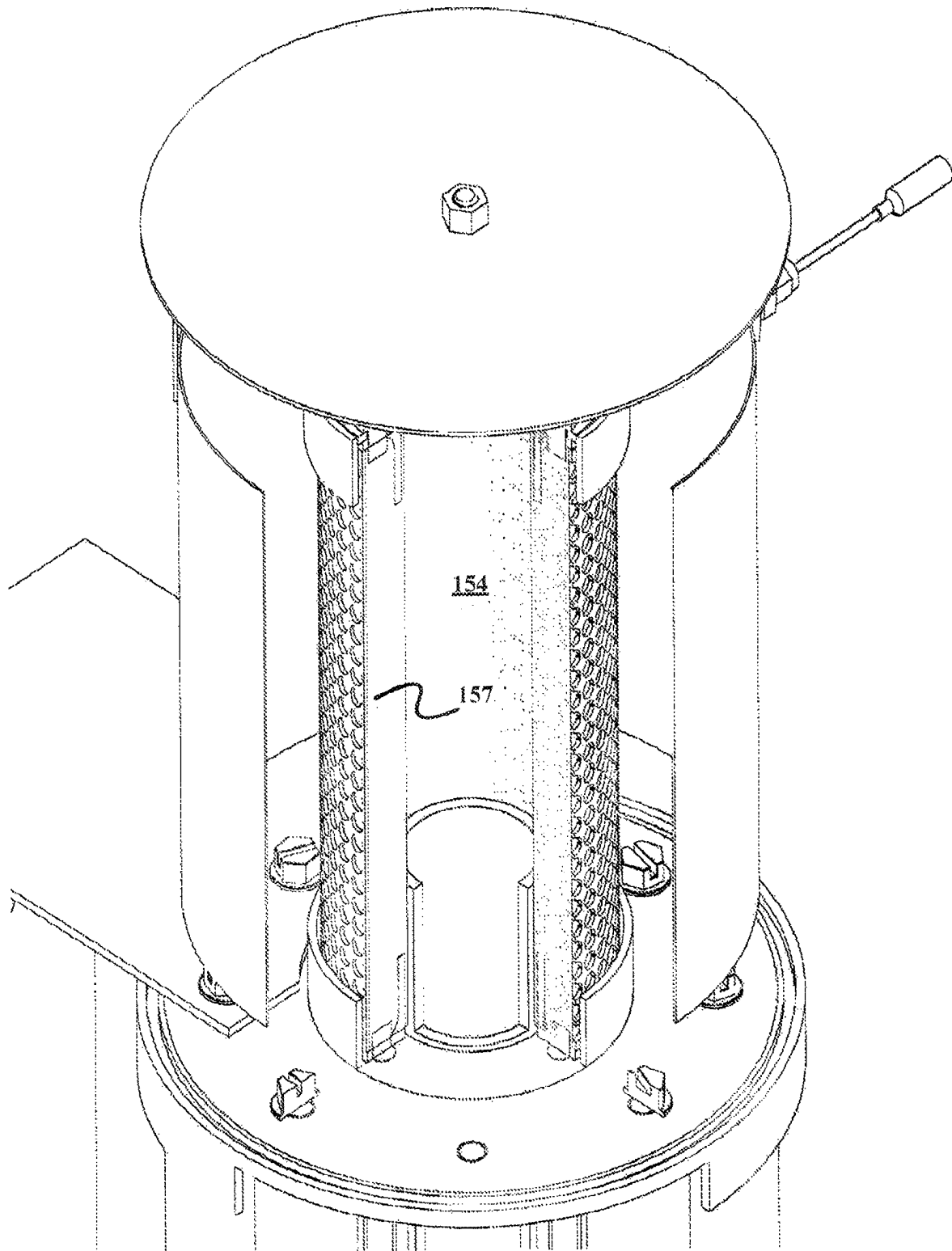


FIG. 6C

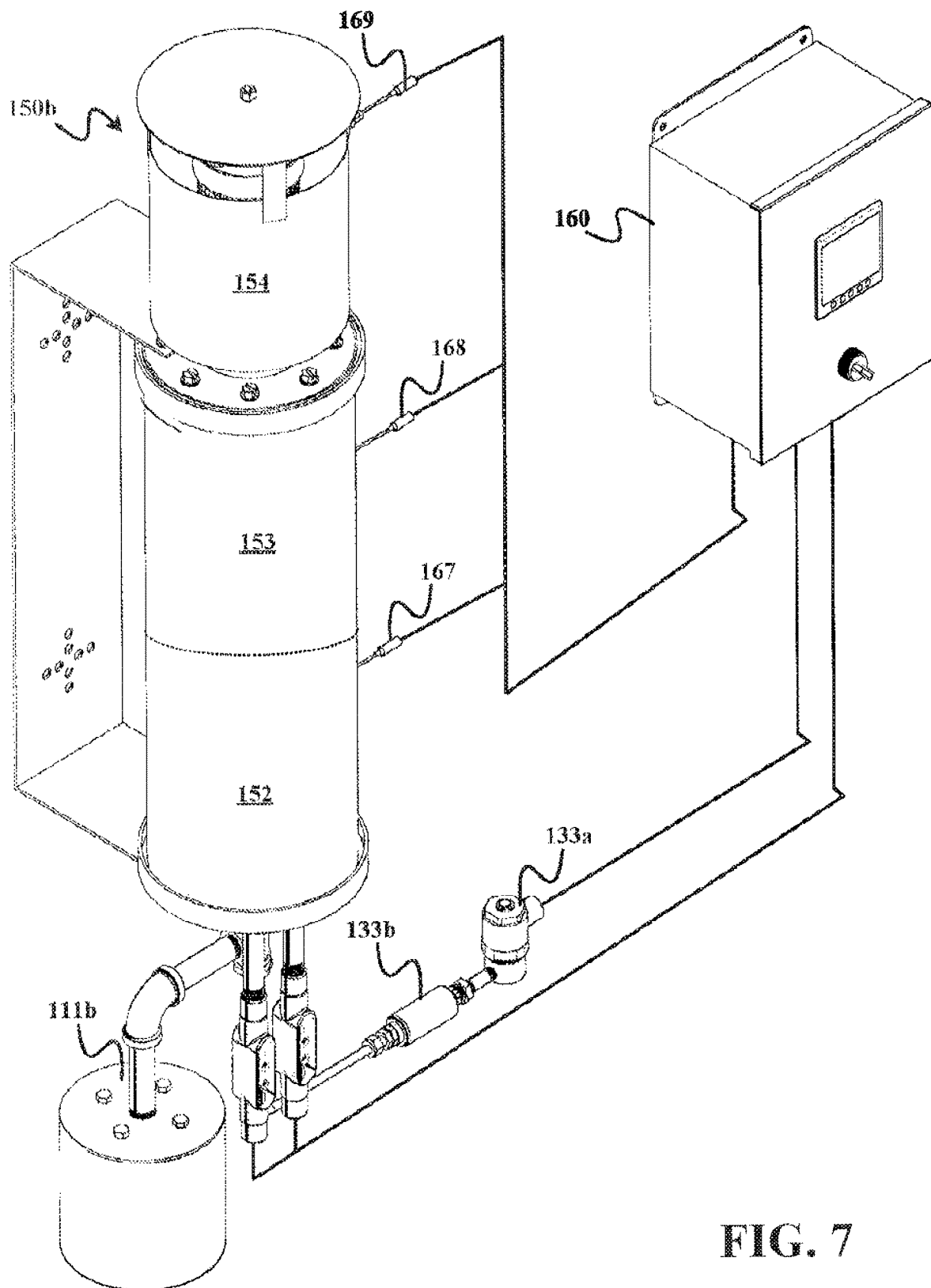
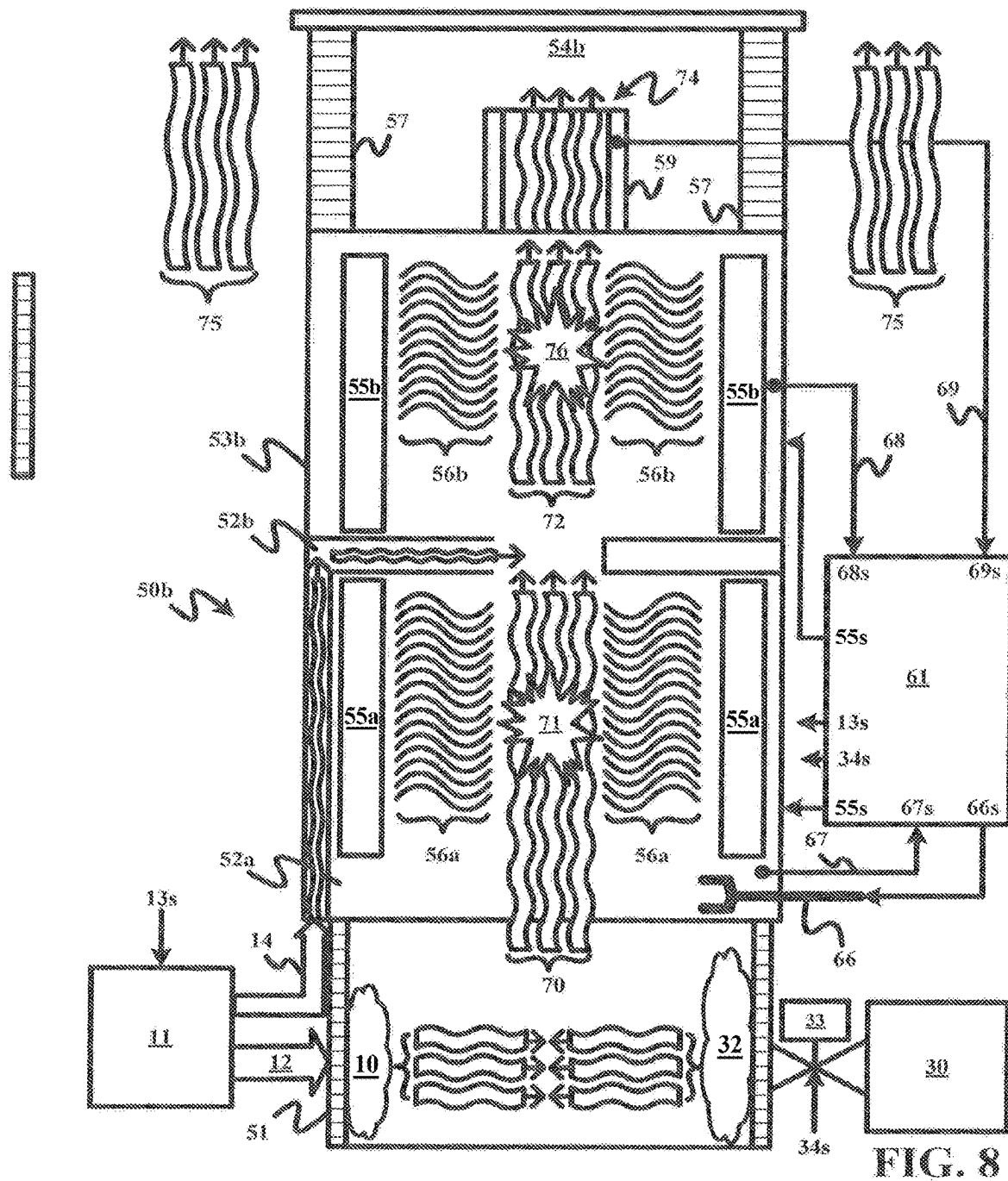


FIG. 7



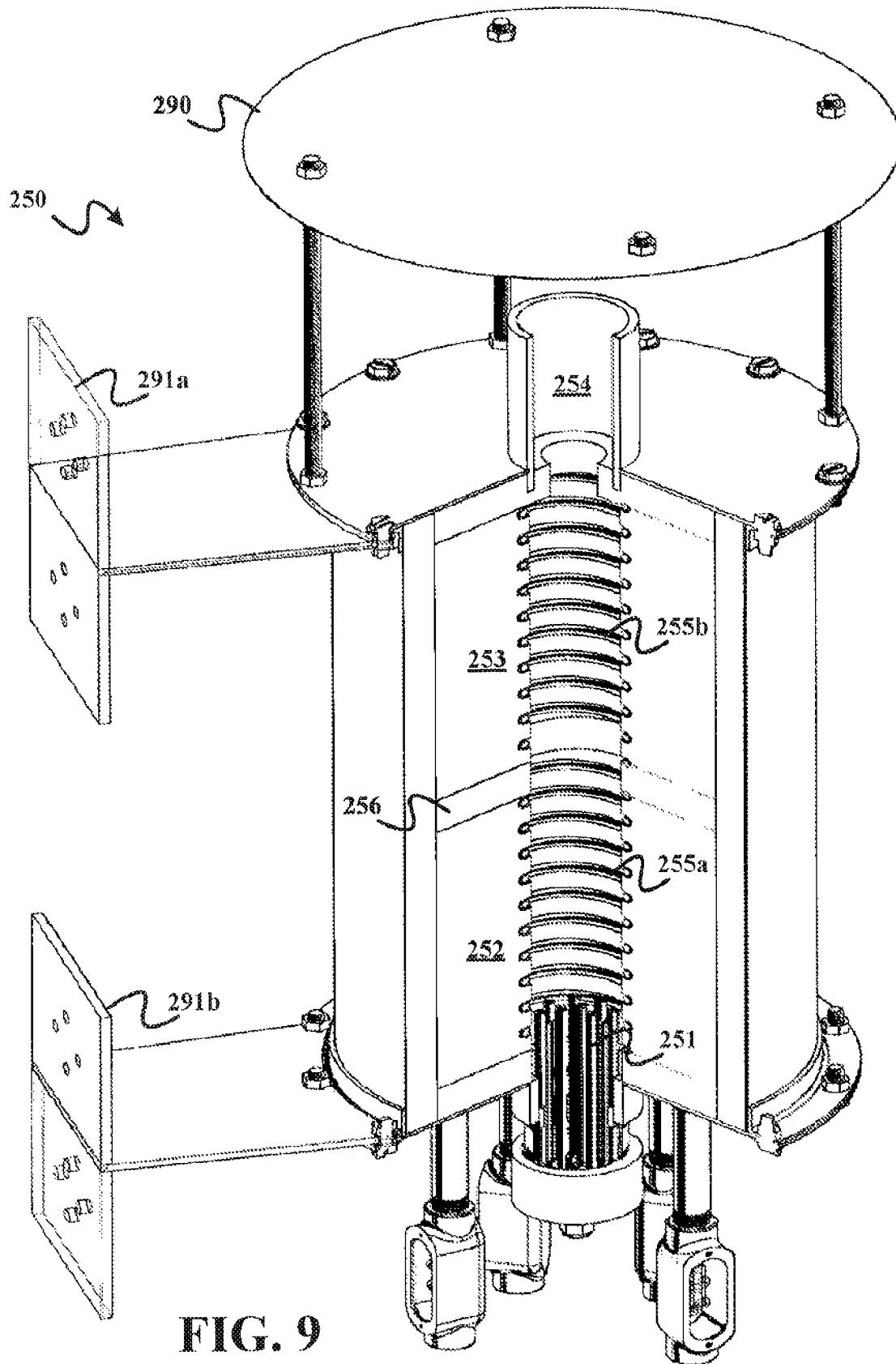
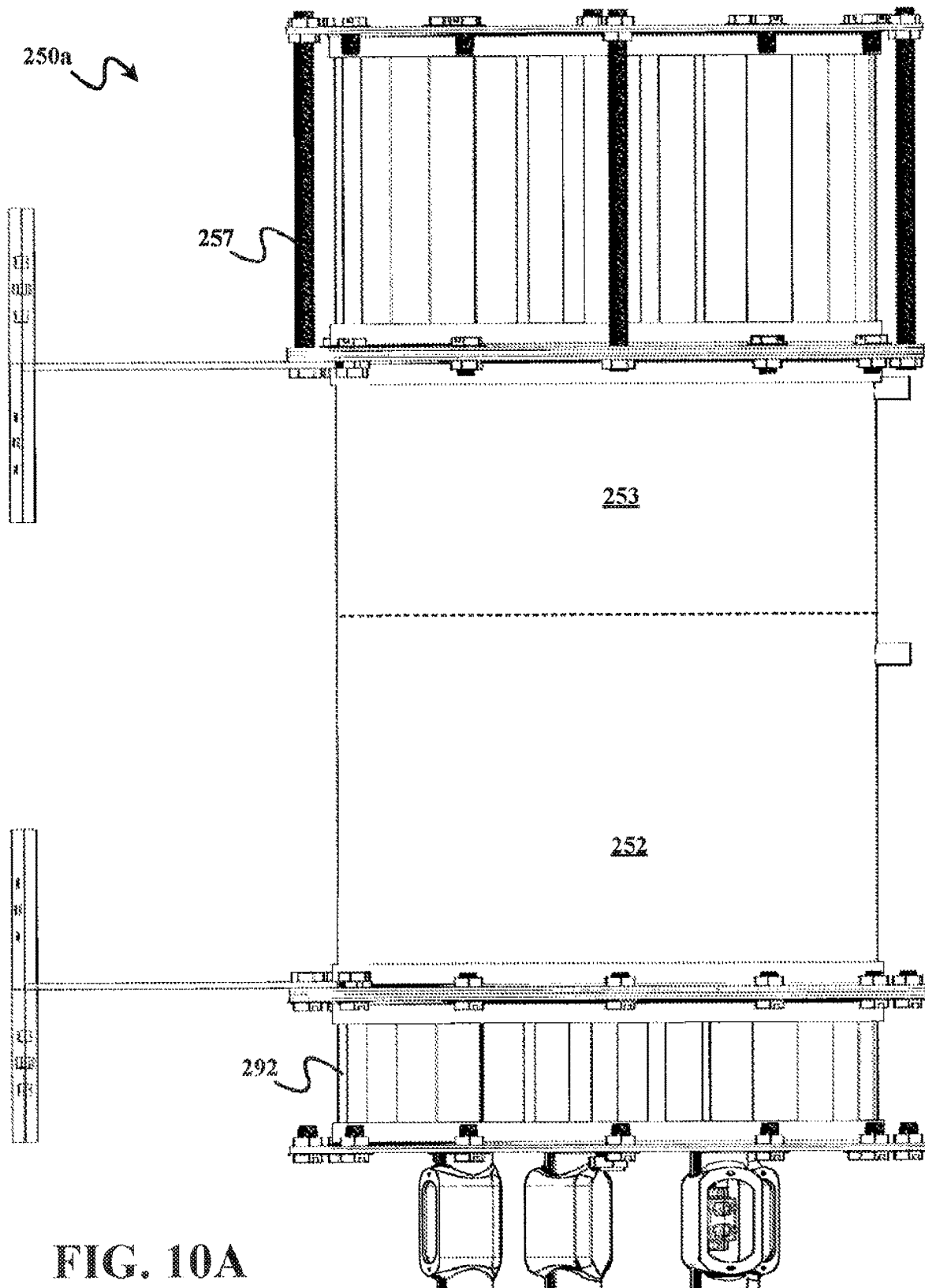
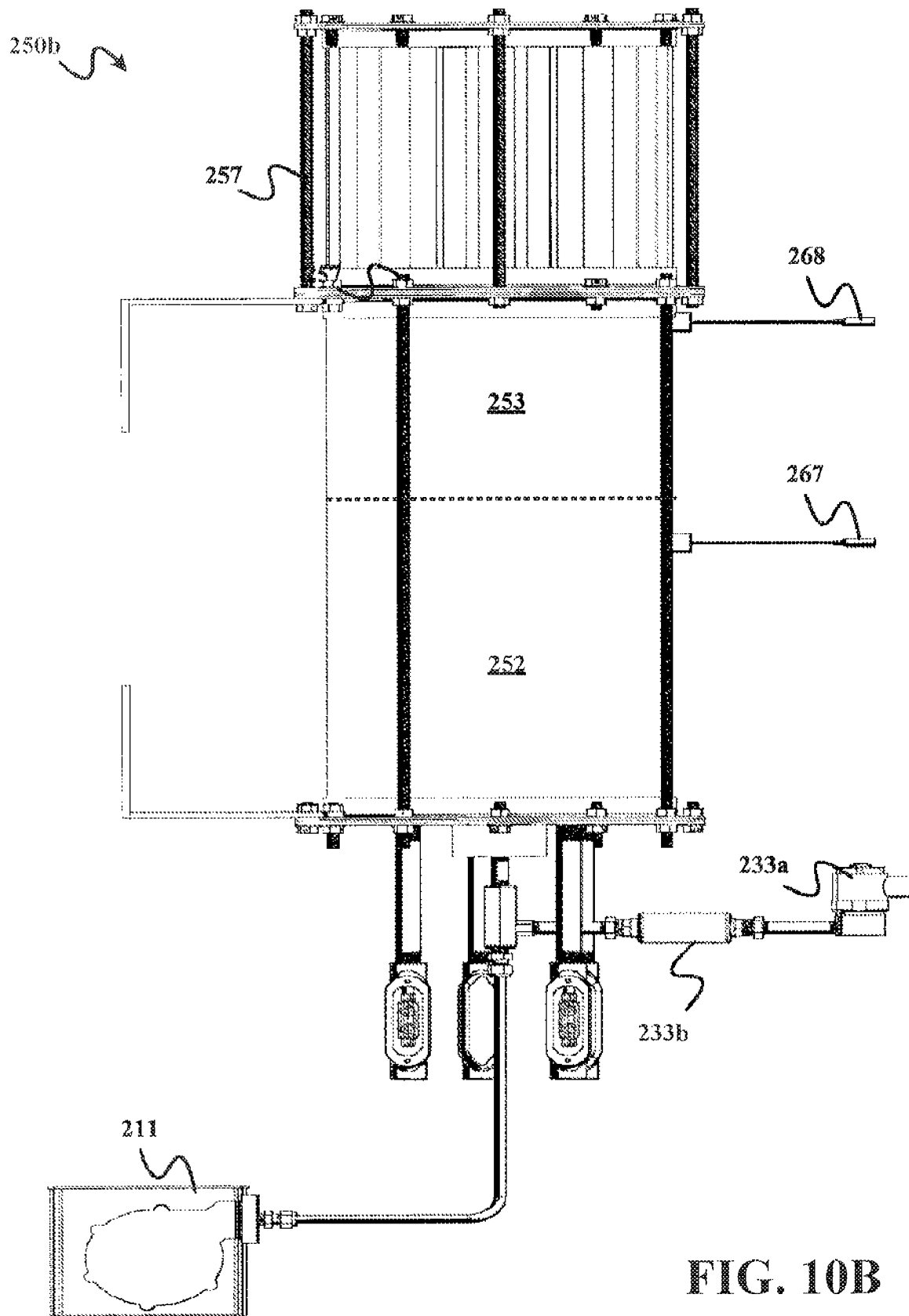


FIG. 9





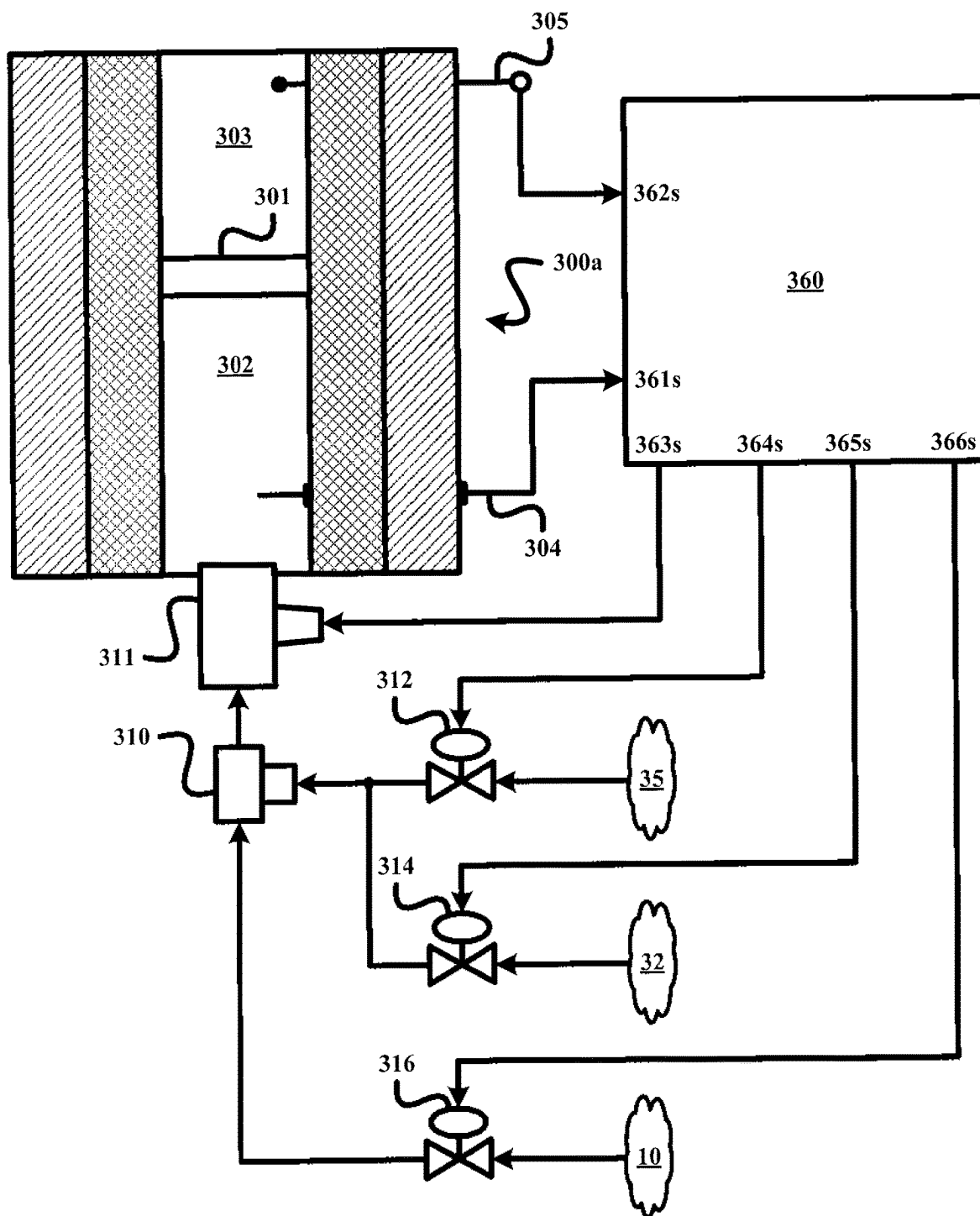


FIG. 11

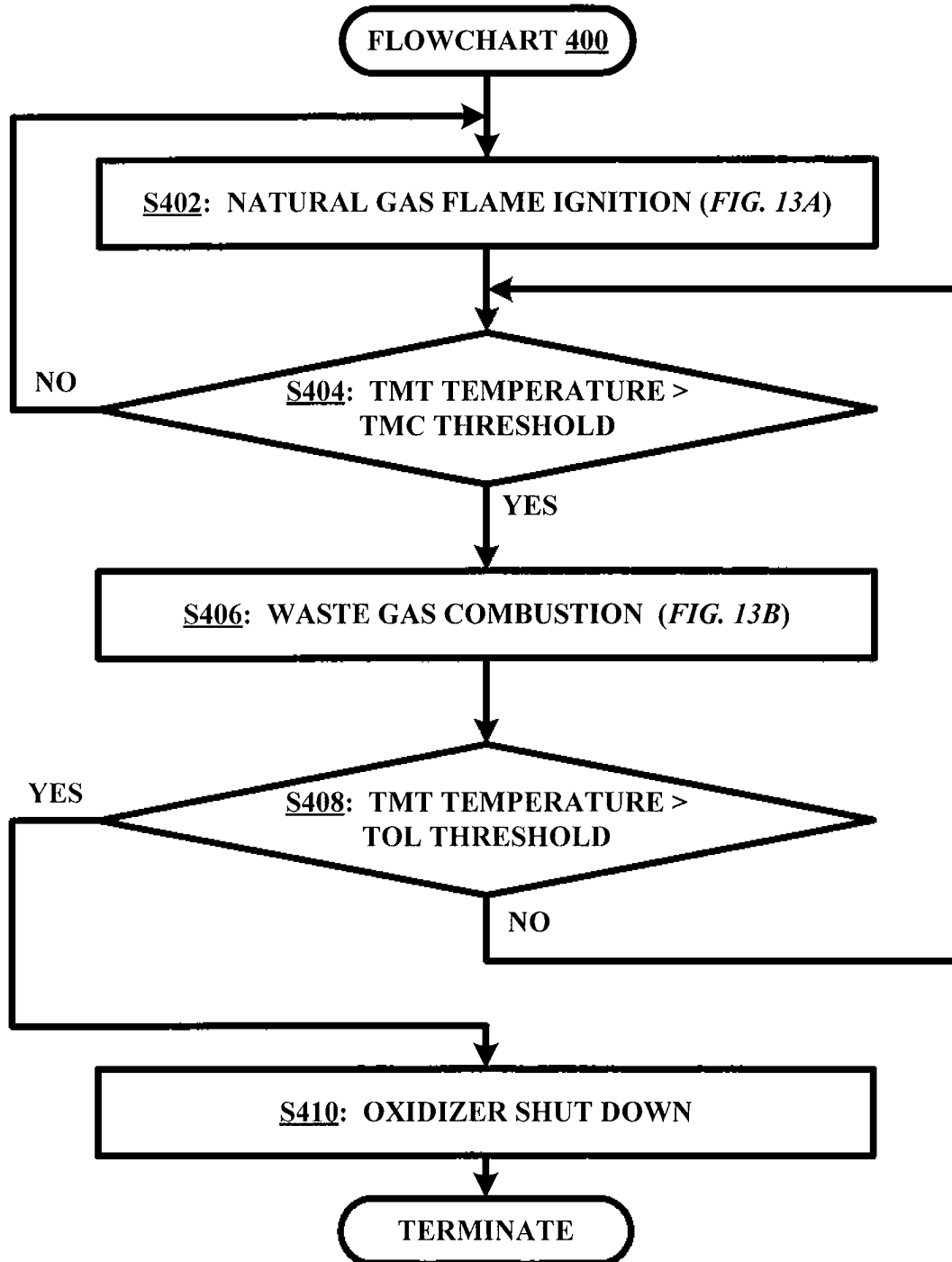


FIG. 12

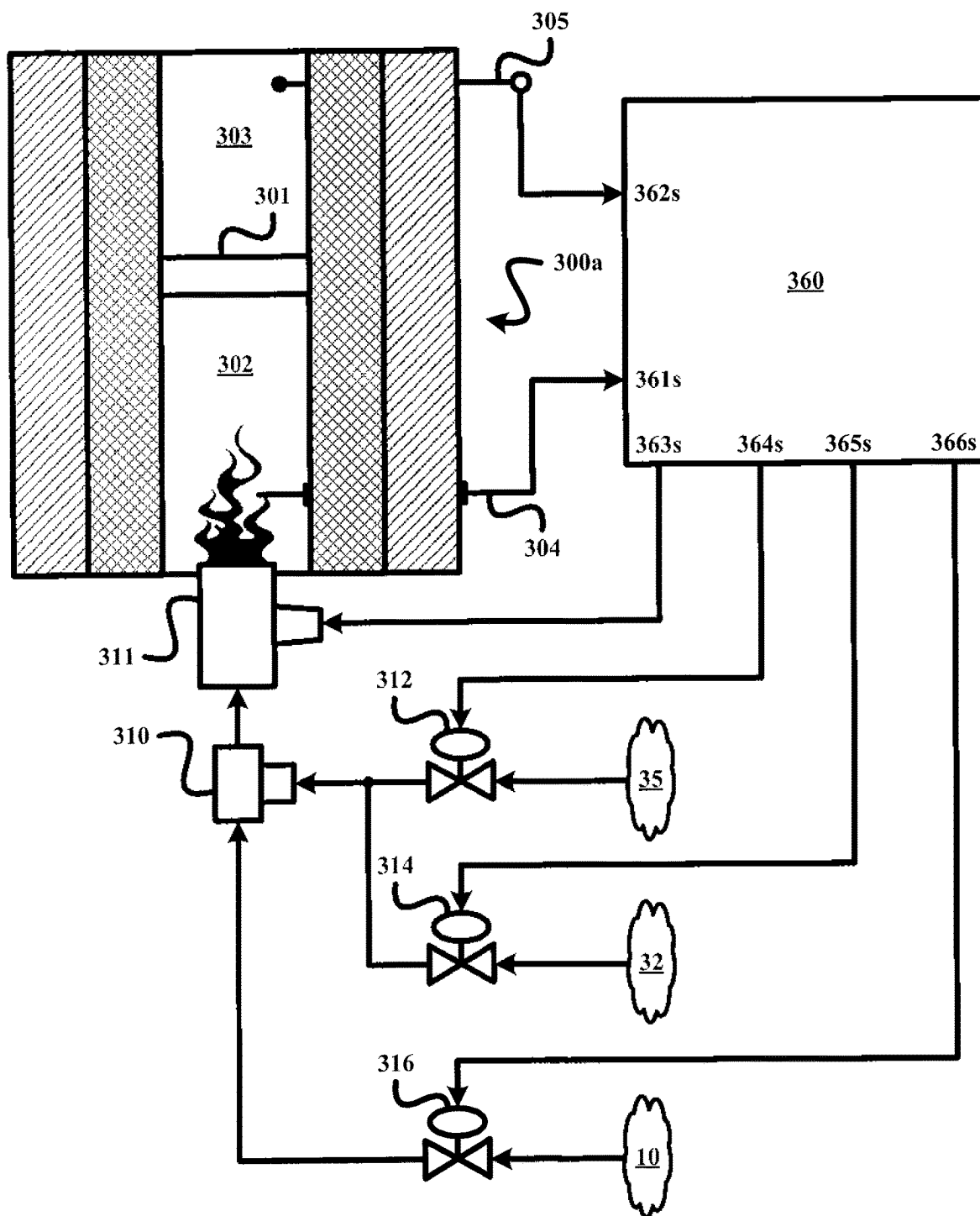


FIG. 13A

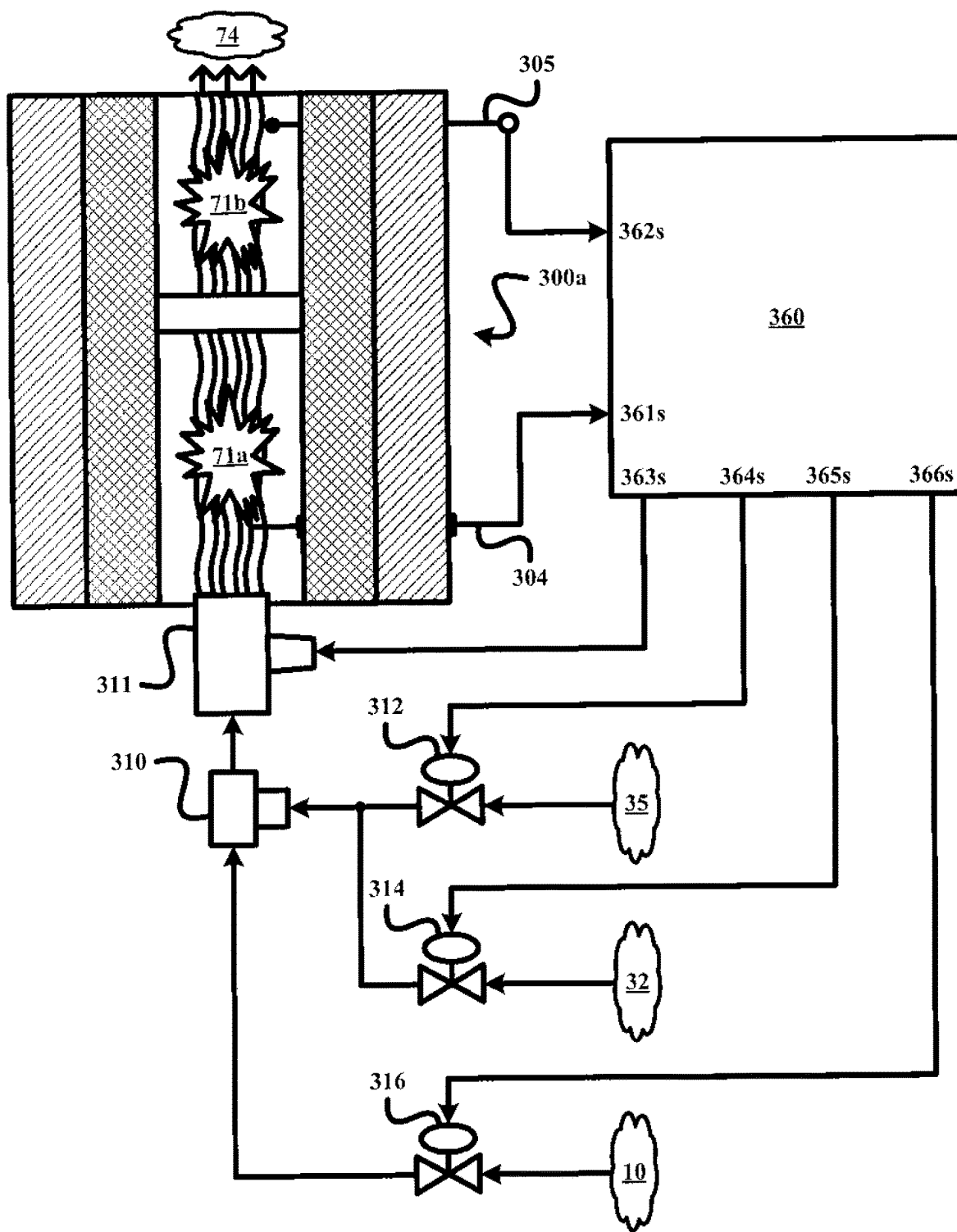


FIG. 13B

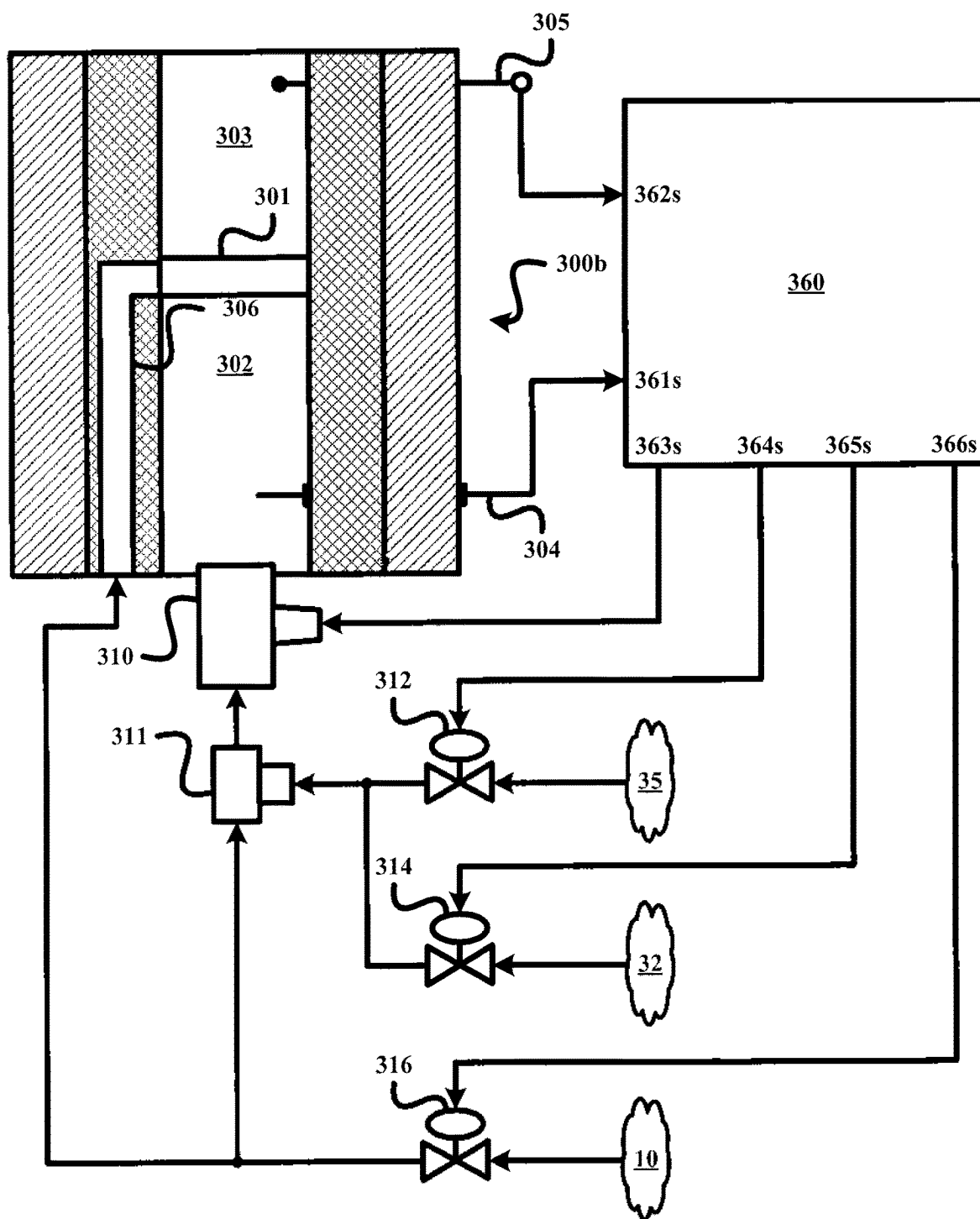


FIG. 14

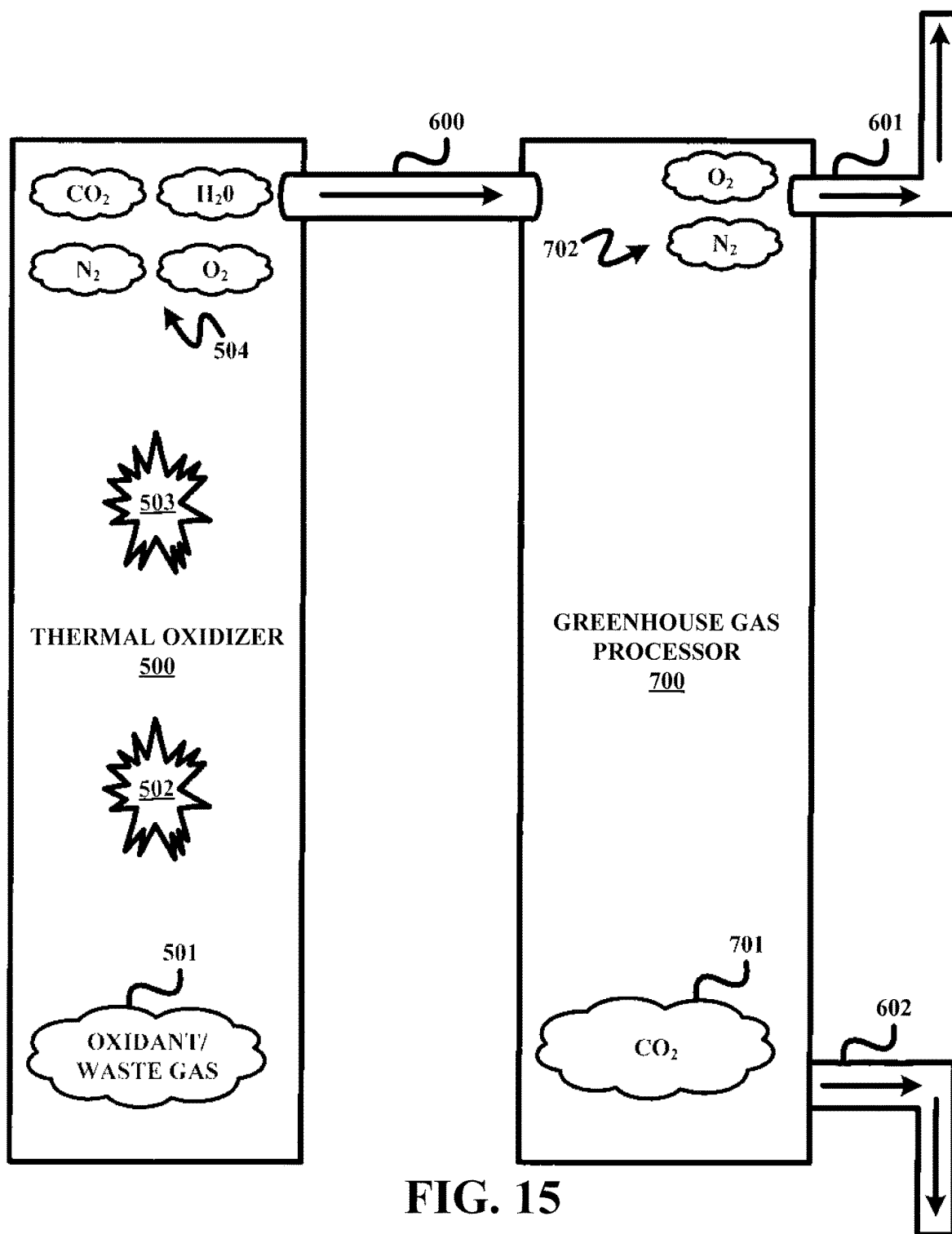


FIG. 15

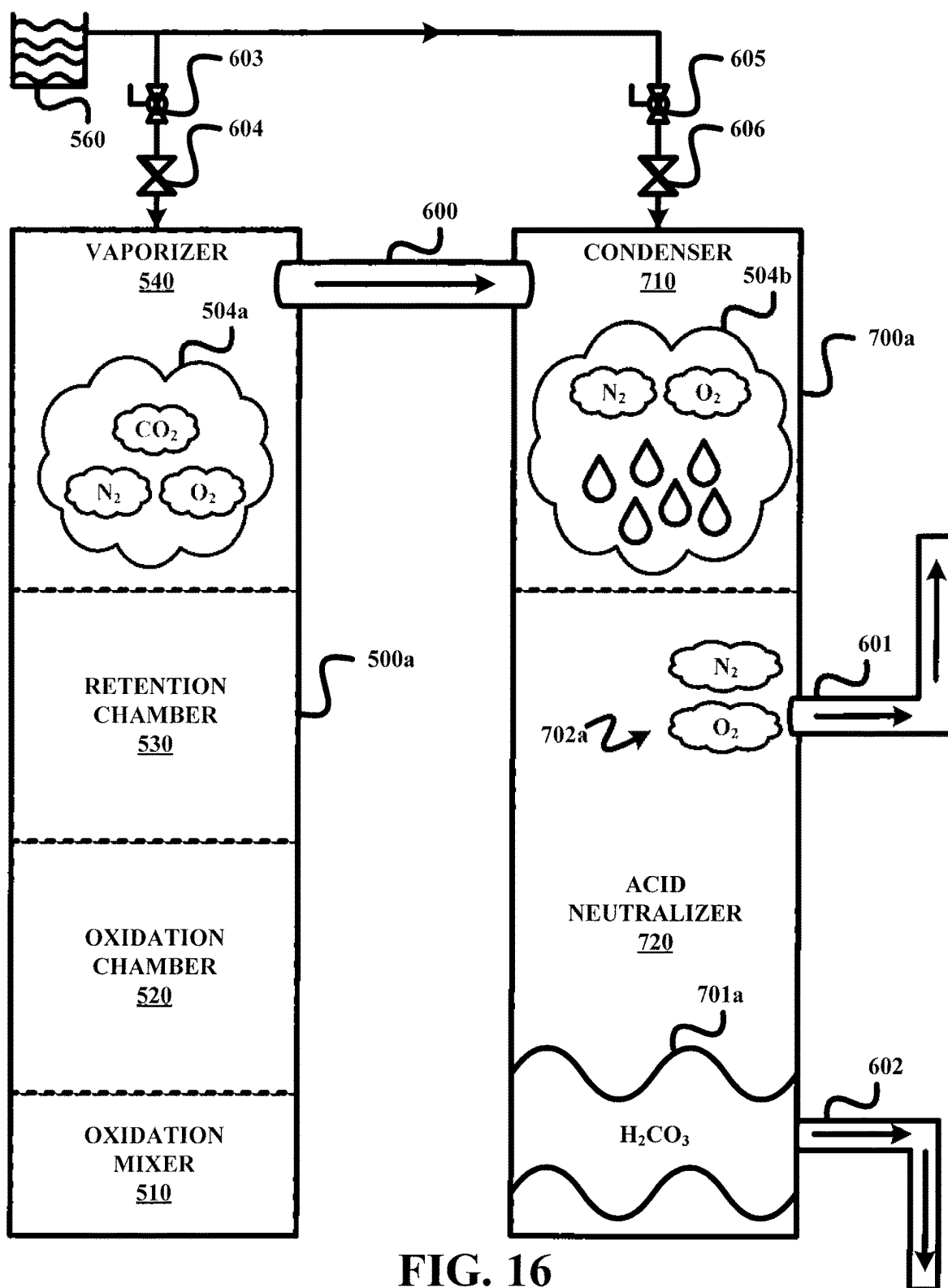


FIG. 16

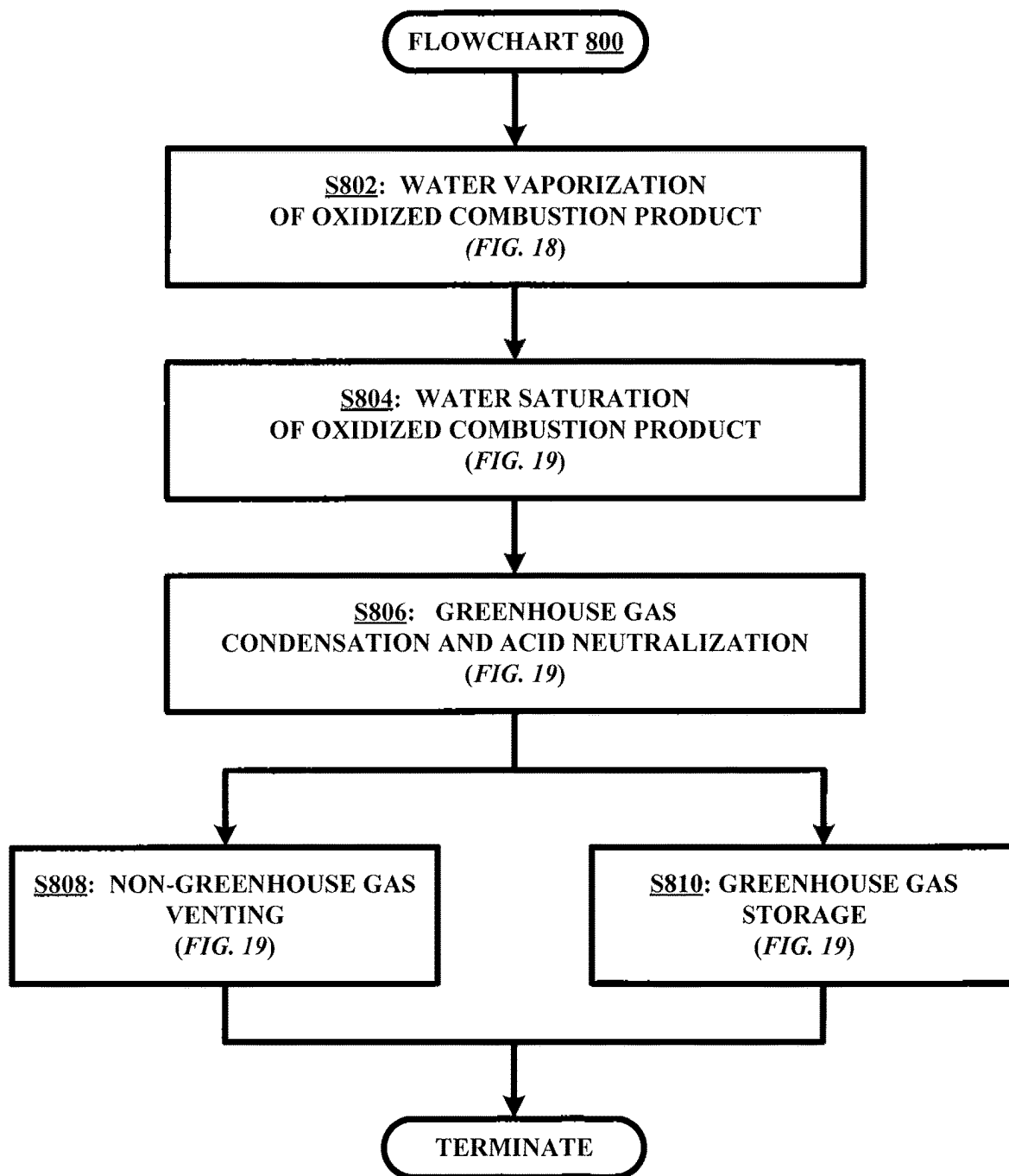


FIG. 17

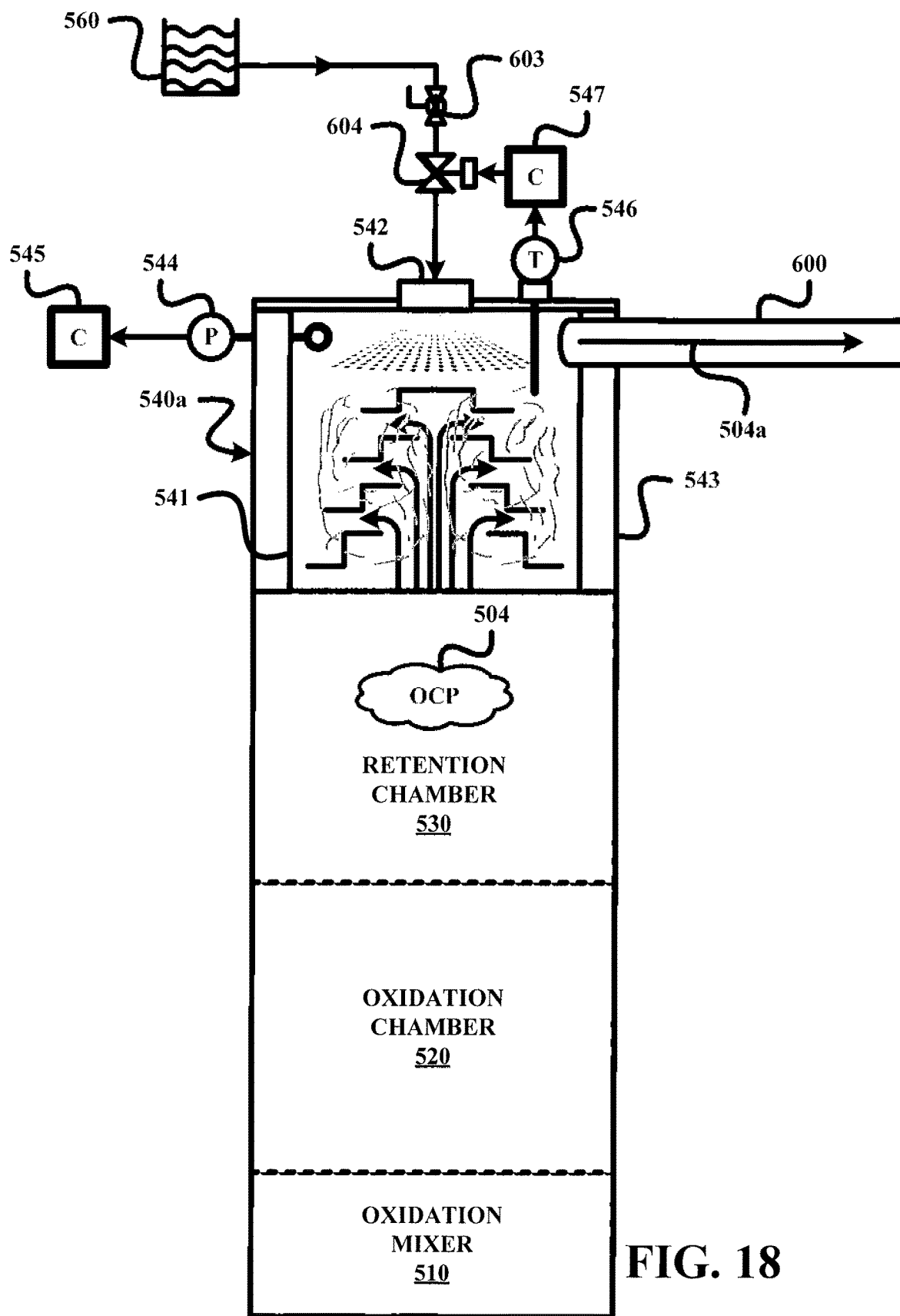


FIG. 18

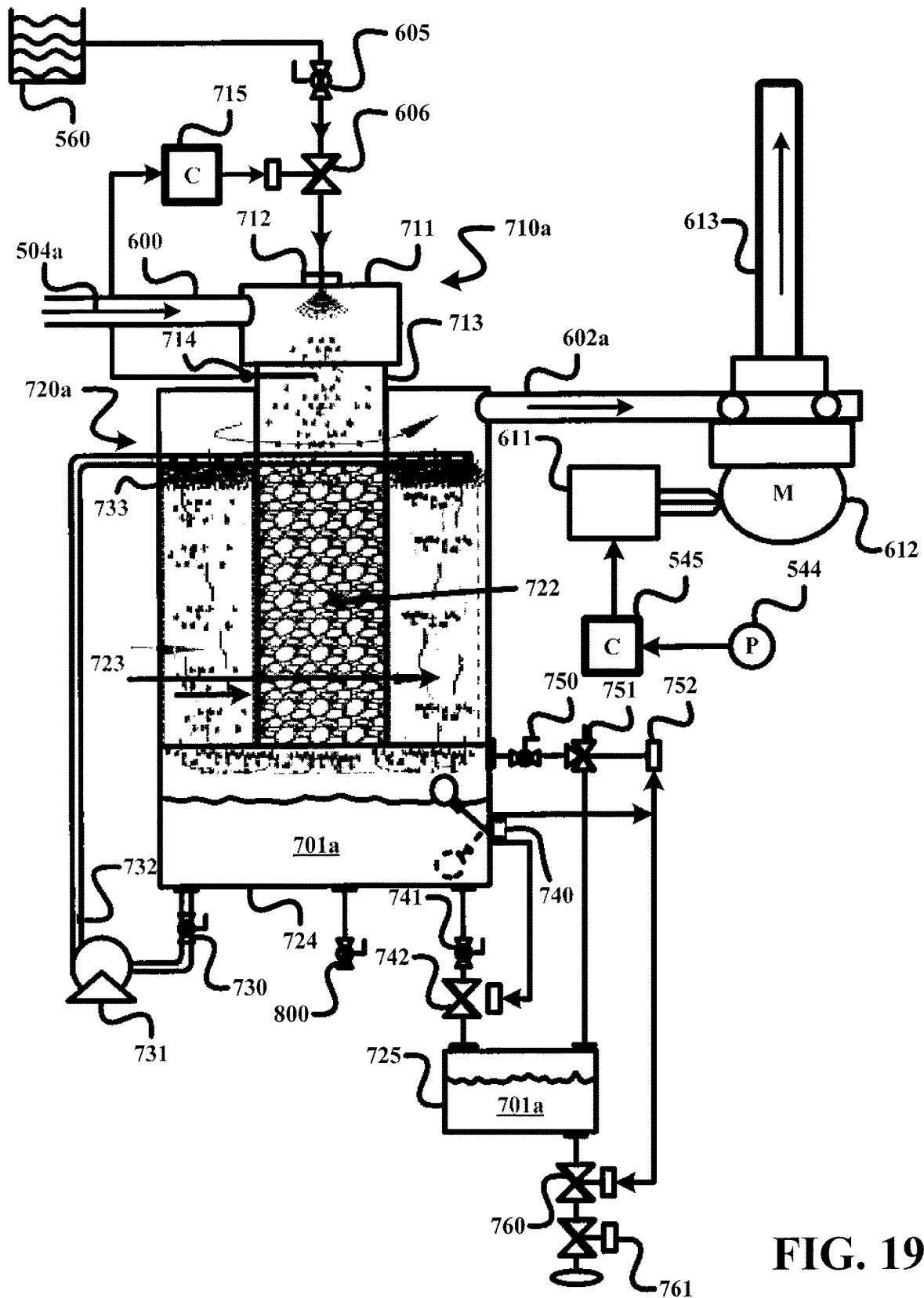


FIG. 19

THERMAL OXIDIZATION SYSTEMS AND METHODS WITH GREENHOUSE GAS CAPTURE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation-in-part of U.S. application Ser. No. 16/313,368 entitled "Thermal Oxidization Systems and Methods," filed Dec. 26, 2018, which is a national stage entry of PCT/US17/39575 entitled "Thermal Oxidization Systems and Methods," filed Jun. 27, 2017, which claims priority to U.S. Provisional Application No. 62/354,991 entitled "Thermal Oxidization Systems and Methods," filed Jun. 27, 2016, (collectively the "Priority Applications"). This present application claims priority and benefit of the Priority Applications. The content of the Priority Applications is incorporated herein by reference.

FIELD OF THE INVENTION

The present disclosure generally relates to a thermal oxidation of waste gas (e.g., toxic release inventory ("TRI") gases, particularly volatile organic compound ("VOC")) into desirable oxidized gases including carbon dioxide gas (CO₂), water vapor (H₂O), nitrogen gas (N₂) and oxygen gas (O₂).

The present disclosure specifically relates to a controlled thermal oxidation of the waste gas involving a regulated combustible mixture of the waste gas and an oxidant (e.g., atmospheric air), a regulated combustion reaction of the combustible mixture into desirable oxidized gases and/or a regulated atmospheric venting of desirable oxidized gases.

BACKGROUND OF THE INVENTION

FIG. 1 illustrates a thermal oxidizer 20 known in the art of the present disclosure. Thermal oxidizer 20 employs a heating chamber 21 and a perforated oxidation reaction ("POR") chamber 24 forming a fluid flow path for a desired oxidation of a waste gas stream 32 (e.g., TRI gases) into oxidized gases 41 (e.g., CO₂, H₂O, N₂ and O₂).

In operation, a heating element 22 within heating chamber 21 is activated to generate heat waves 23 for heating waste gas stream 32 as waste gas stream 32 is feed from a waste gas source 30 via an inlet conduit 31 through heating chamber 21 into POR chamber 24 and as an oxidant 10 (e.g., atmospheric air) flows into POR chamber 24. The heating of waste gas stream 32 is intended to facilitate a combustible oxidation 40 of waste gas stream 32 within POR chamber 24 into oxidized gases 41. However, there are design flaws with thermal oxidizer 20 that impedes a combustible oxidation 40 of waste gas stream 32 within POR chamber 24 into oxidized gases 41.

First, thermal oxidizer 20 fails to regulate a combustible mixture of waste gas stream 32 and oxidant 10 within a flammable range between an upper explosive limit ("UEL") and a lower explosive level ("LEL") as known in the art of the present disclosure, particularly when waste gas stream 32 enters heating chamber 21 at a concentration below the LEL for waste gas stream 32 (i.e., waste gas is too lean). Specifically, a simultaneous suction of oxidant 10 into POR chamber 24 and venting of any gases within POR chamber 24 impedes a sufficient flow of oxidant 10 into POR chamber 24 to ensure a combustible mixture of waste gas stream 32 and oxidant 10 within the flammable range. Consequently, while the combustible mixture of waste gas stream 32 and

oxidant 10 may be within the flammable range upon a power-on of thermal oxidizer 20, the mixture of waste gas stream 32 and oxidant 10 will eventually become too "rich" for combustible oxidation 40 within POR chamber 24. As a result, an incomplete combustion facilitates a buildup of high concentrations of undesirable gases (e.g., carbon monoxide (CO)) and soot within POR chamber 24.

Second, thermal oxidizer 20 fails to regulate a combustion reaction of waste gas stream 32 within POR chamber 24. Specifically, thermal oxidizer 20 does not regulate the heating of heating element 22. Consequently, additional undesirable gases (e.g., nitrogen oxides (NO) and (NO₂)) (not shown) may form within POR chamber 24 if a temperature of heating waves 23 is too high. Moreover, even if the heating of heating element 22 was regulated, POR chamber 24 is configured and sized for an instantaneous combustible oxidation 40 of waste gas stream 32 without a sufficient retention time for a combustible mixture of waste gas stream 32 and oxidant 10, if any, to convert to desirable oxidized gases 41 (e.g., CO₂, H₂O, N₂ and O₂).

The inventions of the present disclosure overcome the drawbacks of prior heating element based thermal oxidizers, particularly thermal oxidizer 20 of FIG. 1.

SUMMARY OF THE INVENTION

One form of the inventions of the present disclosure is a thermal oxidizer employing an oxidation mixer, an oxidation chamber, a retention chamber and a heat dissipater forming a fluid flow path for thermal oxidation of a waste gas (e.g., toxic release inventory ("TRI") gases, particularly volatile organic compound ("VOC")).

In operation, the oxidation mixer facilitates a combustible mixture of the waste gas and an oxidant (e.g., atmospheric air) into a combustible waste gas stream.

For purposes of the inventions of the present disclosure, the term "combustible waste gas stream" broadly encompasses any stream of gas including molecules combinable with oxygen or combust resulting in heat and light, and excludes totally inert gases.

The oxidation mixer may be any form of any oxidation mixer as known in the art of the present disclosure and hereinafter conceived including, but not limited to, a venturi or distributed air-gas mixer.

The oxidation mixer may be fed the waste gas via any type of waste gas feeding mechanism known in the art of the present disclosure and hereinafter conceived including, but not limited to, (1) an aeration nozzle, (2) an aeration nozzle and back flow preventer and (3) an aeration nozzle, back flow preventer and blocking valve.

The oxidation mixer may be fed the oxidant via any type of oxidant feeding mechanism as known in the art of the present disclosure and hereinafter conceived including, but not limited to, (1) an open air inlet involving an inductive air/waste gas flow optionally providing flash back protection, (2) a force modulation air blower with a mixing "T" or (3) a forced modulation air pump and mixing "T".

A heating element or a gas burner of the oxidation chamber generate heat waves facilitating a primary combustion reaction of the combustible waste gas stream flowing from the oxidation mixer to the oxidation chamber.

For purposes of the inventions of the present disclosure, (1) the term "heating element" broadly encompasses any element for converting electricity into heat through the process of Joule/ohmic/resistive, inductive or other means of electrical heating,

3

(2) the term “gas burner” broadly encompasses an element for igniting a natural gas stream,

(3) the term “primary combustion reaction” broadly encompasses an oxidation of the combustible waste gas stream flowing from the oxidation mixer into the oxidation chamber involving a partial combustion of the combustible waste gas stream within the oxidation chamber resulting in an oxygenated waste gas stream, and

(4) the term “oxygenated waste gas stream” broadly encompasses a partial combustion of the combustible waste gas stream in many forms including, but not limited to, CO+other hydrocarbon compounds+CO₂+H₂O+N₂+excess O₂.

The oxygenated waste gas stream flows from the oxidation chamber into the retention chamber whereby the retention chamber facilitates a secondary combustion reaction as needed of the oxygenated waste gas stream into oxidized gases. For purposes of the inventions of the present disclosure, the term “secondary combustion reaction” encompasses a complete oxidation of the oxygenated waste gas stream flowing from the oxidation chamber to the retention chamber involving a conversion of the oxygenated waste gas stream into oxidized gases (e.g., CO₂, H₂O, N₂ and O₂). The retention chamber may also include an additional heating element to facilitate the complete oxidation of oxygenated waste gas stream into oxidized gases.

Concurrently or alternatively, the oxidation chamber may further include an supplemental oxidant inlet for mixing additional oxidant to the oxygenated waste gas stream into a combustible oxygenated waste gas stream flowing into the retention chamber.

The heat dissipater facilitates an atmospheric venting of the oxidized gases flowing into the heat dissipater from the retention chamber.

Another form of inventions of the present disclosure is a thermal oxidation system including a thermal oxidizer of the present disclosure and a greenhouse gas processor. For purposes of the inventions of the present disclosure, the term “greenhouse gas processor” broadly encompasses any machine structurally configured to extract greenhouse gas(es) within oxidized gases.

For this thermal oxidation system, a heat dissipator of the thermal oxidizer heat dissipates the flow of oxidized gases within the heat dissipator and communicates the oxidized gases at a reduced temperature to the greenhouse gas processor.

In operation, the greenhouse gas processor extracts greenhouse gas(es) from within the oxidized gases, and may further vent non-greenhouse gas(es) within the oxidized gases.

In extracting the greenhouse gas(es) from within the oxidized gases, the greenhouse gas processor may liquid condensate the greenhouse gas(es), acid neutralize the condensation of the greenhouse gas(es) and capture the acid neutralized condensation of the greenhouse gas(es).

The foregoing forms and other forms of the present invention as well as various features and advantages of the present invention will become further apparent from the following detailed description of various embodiments of the present invention read in conjunction with the accompanying drawings. The detailed description and drawings are merely illustrative of the present invention rather than limiting, the scope of the present invention being defined by the appended claims and equivalents thereof.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a block diagram of an exemplary thermal oxidizer as known in the art.

4

FIG. 2 illustrates a block diagram of a first exemplary chamber embodiment of a thermal oxidization system in accordance with the inventive principles of the present disclosure.

FIG. 3 illustrates a block diagram of an exemplary embodiment of a control system in accordance with the inventive principles of the present disclosure.

FIGS. 4A-4C illustrate a flowchart representative of an exemplary embodiment of a thermal oxidization method in accordance with the inventive principles of the present disclosure.

FIGS. 5A and 5B illustrate perspective views of a first exemplary embodiment of the thermal oxidization system of FIG. 2 in accordance with the inventive principles of the present disclosure.

FIGS. 6A-6C illustrate exploded views of the thermal oxidization system of FIGS. 5A and 5B.

FIG. 7 illustrates a perspective view of a second exemplary embodiment of the thermal oxidization system of FIG. 2 in accordance with the inventive principles of the present disclosure.

FIG. 8 illustrates a block diagram of a second exemplary embodiment of a thermal oxidization system in accordance with the inventive principles of the present disclosure.

FIG. 9 illustrates a perspective view of an exemplary embodiment of the thermal oxidizer of FIG. 8 in accordance with the inventive principles of the present disclosure.

FIG. 10A illustrates an exemplary atmospheric embodiment of the thermal oxidizer of FIG. 9.

FIG. 10B illustrates an exemplary forced air embodiment of the thermal oxidizer of FIG. 9.

FIG. 11 illustrates a first exemplary embodiment of a gas fired thermal oxidizer in accordance with the inventive principles of the present disclosure.

FIG. 12 illustrates a flowchart representative of an exemplary embodiment of a gas fired thermal oxidization method in accordance with the inventive principles of the present disclosure.

FIG. 13A illustrates an exemplary ignition phase of the flowchart of FIG. 12.

FIG. 13B illustrates an exemplary combustion phase of the flowchart of FIG. 12.

FIG. 14 illustrates a second exemplary embodiment of a gas fired thermal oxidizer in accordance with the inventive principles of the present disclosure.

FIG. 15 illustrates an exemplary embodiment of greenhouse gas processor based thermal oxidation system of the present disclosure in accordance with the inventive principles of the present disclosure.

FIG. 16 illustrates an exemplary embodiment of the greenhouse gas processor based thermal oxidation system of FIG. 15 in accordance with the inventive principles of the present disclosure.

FIG. 17 illustrates a flowchart representative of an exemplary embodiment of greenhouse gas processing method in accordance with the inventive principles of the present disclosure.

FIG. 18 illustrates an exemplary embodiment of the vaporizing heat dissipator of FIG. 17 in accordance with the inventive principles of the present disclosure.

FIG. 19 illustrates exemplary embodiments of a condenser and an acid gas neutralizer of FIG. 1 in accordance with the inventive principles of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

To facilitate an understanding of inventions of the present disclosure, the following description of FIGS. 2-4 teach

basic inventive principles of thermal oxidization systems and thermal oxidization methods of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of thermal oxidization systems and thermal oxidization methods of the present disclosure.

Referring to FIG. 2, a thermal oxidization system of the present disclosure incorporates a thermal oxidizer 50a employing an oxidation mixer 51, an oxidation chamber 52a, a retention chamber 53a and a heat dissipater 54a forming a fluid flow path for oxidation of a waste gas stream 32 (e.g., TRI gases) into oxidized combustion products 74 (e.g., oxidized gases 73 including CO₂, H₂O, N₂ and O₂).

Oxidation mixer 51 is structurally configured for implementing a combustible mixture of an oxidant 10 (e.g., atmospheric air) supplied by oxidant supply 11 (e.g., a blower or a pump) via a supply line 12 and of a waste gas stream 32 (e.g., TRI gases) supplied by a waste gas source 30 as controlled via a control flow conduit 33 (e.g., a solenoid valve and a flame arrestor in series coupling waste gas source 30 to oxidation mixer 51) into a combustible waste gas stream 70.

In one embodiment, oxidation mixer 51 is a venturi air-gas mixer whereby turbulent fluid flows of oxidant 10 and waste gas stream 32 into the venturi air-gas mixer are controlled via oxidant supply 11 and control flow conduit 33 to ensure combustible waste gas stream 70 attains proportional concentrations of oxidant 10 and waste gas stream 32 within a flammable range (e.g., 11.5:1 oxidant 10 to waste gas stream 32). Additionally, oxidation mixer 51 may be equipped with a nozzle (not shown) for regulating a feeding of combustible waste gas stream 70 into oxidation chamber 52a whereby the nozzle may be structurally configured to generate more turbulence to combustible waste gas stream 70.

Oxidation chamber 52a is structurally configured for implementing a primary combustion reaction 71 therein of combustible waste gas stream 70 into an oxygenated waste gas stream 72 via a controlled emission of heat waves 56 by a heating element 55. In one embodiment, oxidation chamber 52a is a refractory ceramic cylinder and heating element 55 is embedded within the walls of the refractory ceramic cylinder.

Optionally, oxidation chamber 52a may further employ a spark igniter 66 for a controlled ignition of combustible waste gas stream 70 at a proximal opening of oxidation chamber 52a. For this embodiment, if oxidation mixer 51 is equipped with a nozzle, then a distal tip of spark igniter 66 may be positioned within or adjacent to the flow of the combustible waste gas stream 70 out of the nozzle into oxidation chamber 52a.

Retention chamber 53a is structurally configured for implementing a retention time for a secondary combustion reaction of oxygenated waste gas stream 72 into heated oxidized combustion products 74 (e.g., oxidized gases 73 including CO₂, H₂O, N₂ and O₂). In one embodiment, retention chamber 53a is a refractory ceramic cylinder integrated with oxidation chamber 52a as shown.

Heat dissipater 54a is structurally configured for implementing a heat exchange of atmosphere air 75 with heated oxidized combustion products 74 to vent cooled oxidized combustion products 74 into the atmosphere. In one embodiment, heat dissipater 54a includes a heat exchanger 57a constructed of stainless steel woven fabric, which has been pleated and rolled into a cylinder shape whereby oxidized

combustion products 74 exits heat exchanger 57a along a length and circumference of the vertical wall of heat exchanger 57a as shown with cooling atmosphere air 75 being directed vertically past the vertical wall of heat exchanger 57a to thereby extract heat from heat dissipater 54a.

Additionally, heat dissipater 54a may be equipped with mesh baffles (e.g., mesh baffles 58b and 58c as shown) axially aligned on a longitudinal axis of the cylindrical heat exchanger 57a to thereby provide a more controlled flow diversion of oxidized combustion products 74 in a direction of vertical wall of heat exchanger 57a as shown in FIG. 2.

Referring to FIG. 3, thermal oxidization system of the present disclosure further incorporates a control system employing an oxidation controller 61, a data logger 63 and a data reporter 65 housed within a control box 60.

Oxidation controller 61 is structurally configured for controlling an operation of thermal oxidizer 50a as will be further described herein in connection with a description of FIGS. 4A-4C.

In one embodiment, oxidation controller 61 is an application specific main board or an application specific integrated circuit for controlling a thermal oxidation application of various inventive principles of the present disclosure as subsequently described herein in connection with FIGS. 4A-4C. The structural configuration of oxidation controller 61 may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for controlling an execution of the thermal oxidation application.

A non-limiting example of oxidation controller 61 is an all-in-one XL4 controller sold by Horner APG of Indianapolis, Ind. that is configured in accordance with the inventive principles of the present disclosure.

Data logger 63 is structurally configured for logging operational data ("OD") 62 transmitted by oxidation controller 61 to data logger 63 via a push or pull operation, or by a monitoring of specific data points of oxidation controller 61 by data logger 63. Operational data 62 includes data informative of an operational status of thermal oxidizer 50a in executing the oxidation of waste gas stream 32.

In one embodiment, data logger 63 is an application specific main board or an application specific integrated circuit for controlling a data logging application of the present disclosure. The structural configuration of data logger 63 may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for executing the data logging application.

A non-limiting example of data logger 63 is an all-in-one XLE controller sold by Horner APG of Indianapolis, Ind. that is configured in accordance with the inventive principles of the present disclosure.

Another non-limiting example of data logger 63 is as an application module configured within oxidation controller 61.

Also in practice, data logger 63 may be omitted and oxidation controller 61 may be configured for executing the data logging application of the present disclosure.

Data reporter **65** is structurally configured for receiving reporting data ("RD") **64** from data logger **63** via a push or pull operation and transmitting reporting data **64** to a network **80** (e.g., a cellular network). Reporting data **64** is operational data **62** processed by data logger **63** into a reporting format associated with data reporter **65**.

In one embodiment, data reporter **65** is an application specific modem for executing a data reporting application of the present disclosure.

A non-limiting example of data logger **66** is a SonicU™ modem sold by SonicU, LLC of Greenfield, Ind. that is configured in accordance with the inventive principles of the present disclosure.

Referring back to FIG. 2, oxidation controller **61** regulates a control of thermal oxidizer **50a** in accordance with a thermal oxidization method of the present disclosure as represented by a flowchart **90** shown in FIGS. 4A-4C. In support of executing the thermal oxidization method:

1. oxidation controller **61** senses and controls a flash point temperature T_{FP} of oxidation chamber **52a** via a connection of thermocouple **67** at a signal input **67s**;
2. oxidation controller **61** senses and controls an oxidation point temperature T_{OP} of retention chamber **53a** via a connection of thermocouple **68** at a signal input **68s**;
3. oxidation controller **61** senses and controls a venting point temperature T_{VP} of heat dissipater **54a** via a connection to thermocouple **69** at a signal input **69s**;
4. oxidation controller **61** controls a feeding rate of oxidant **10** by oxidant supply **11** via a connection of a signal output **13s** to oxidant supply **11a**;
5. oxidation controller **61** controls a feeding rate of waste gas stream **32** by waste gas source **30** via a connection of a signal output **34s** to control flow conduit **33**;
6. oxidation controller **61** controls an activation and a deactivation of heating element **55** via a signal output **55s**; and
7. oxidation controller **61** controls an activation and a deactivation of spark igniter **66** via a signal output **66s**;

Referring to FIGS. 2 and 4A, prior to be oxidation controller **61** being powered-on, oxidant supply **11** is deactivated and control flow conduit **33** is closed.

Upon oxidation controller **61** being powered-on, oxidation controller **61** executes flowchart **90**, and initiates a transmission of a point monitoring of flash point temperature T_{FP} , oxidation point temperature T_{OP} and venting point temperature T_{VP} to data logger **63** for recording purposes.

As to the execution of flowchart **90** by oxidation controller **61**, a stage **S92** of flowchart **90** encompasses a thermocouple validity check ("TVC") by oxidation controller **61** concurrently or sequentially involving:

1. a comparison by oxidation controller **61** of flash point temperature T_{FP} relative to a flash thermocouple validity threshold ("FTVT");
2. a comparison by oxidation controller **61** of oxidation point temperature T_{OP} relative to an oxidation thermocouple validity threshold ("OTVT"); and
3. a comparison by oxidation controller **61** of venting point temperature T_{VP} relative to a venting thermocouple validity threshold ("VTVT").

In practice, to test the validity of the thermocouples, the thermocouple validity thresholds should be set at an anticipated maximum room temperature (e.g., 22° C.) or to an anticipated maximum local outdoor air temperature.

For a TVC cycle of stage **S92**, if flash point temperature T_{FP} is more than flash thermocouple validity threshold FTVT, OR if oxidation point temperature T_{OP} is more than oxidation thermocouple validity threshold OTVT, OR if

venting point temperature T_{VP} is more than venting thermocouple validity threshold VTVT, then oxidation controller **61** is powered down for a maintenance inspection of the thermocouple(s) and powered on again after a successful repair/replacement, if any, of the thermocouple(s).

Otherwise, if flash point temperature T_{FP} is less than or equal to flash thermocouple validity threshold FTVT, AND if oxidation point temperature T_{OP} is less than or equal to oxidation thermocouple validity threshold OTVT, AND if venting point temperature T_{VP} is less than or equal to venting thermocouple validity threshold VTVT, then oxidation controller **61** proceeds to a stage **S94** of flowchart **90** encompassing a thermal oxidation limit check ("TOLC") by oxidation controller **61** concurrently or sequentially involving:

1. a comparison by oxidation controller **61** of flash point temperature T_{FP} relative to a flash thermal limit threshold ("FTLT");
2. a comparison by oxidation controller **61** of oxidation point temperature T_{OP} relative to an oxidation thermal limit threshold ("OTLT"); and
3. a comparison by oxidation controller **61** of venting point temperature T_{VP} relative to a venting thermal limit threshold ("VTLT").

In practice, to ensure a safe heat activation of oxidation chamber **52a**, the thermal limit thresholds should represent a predefined maximum temperature of oxidation chamber **52a**, retention chamber **53a** and heat dissipater **54a** prior to the heat activation of thermal oxidizer **50a** (e.g., 300° F. for all thermocouple limit thresholds).

For a TOLC cycle of stage **S94**, if flash point temperature T_{FP} is more than flash thermal limit threshold FTLT, OR if oxidation point temperature T_{OP} is more than oxidation thermal limit threshold OTLT, OR if venting point temperature T_{VP} is more than venting thermal limit threshold VTLT, then oxidation controller **61** returns to stage **S94** to execute a new TOLC cycle.

Otherwise, if flash point temperature T_{FP} is less than or equal to flash thermal limit threshold FTLT, AND if oxidation point temperature T_{OP} is less than or equal to oxidation thermal limit threshold OTLT, AND if venting point temperature T_{VP} is less than or equal to venting thermal limit threshold VTLT, then oxidation controller **61** proceeds to a stage **S96** of flowchart **90** to execute a pre-oxidation stage of the thermal oxidation of waste gas stream **32** sequentially involving:

1. with control flow conduit **33** being closed via valve control signal VC (e.g., a binary open-close signal) and heating element **55** being deactivated via heat regulation signal HR (e.g., an AC current), oxidation controller **61** controlling a gaseous purge of thermal oxidizer **50a** by activating oxidant supply **11** via a supply control signal SC (e.g., a digital amplitude signal) for a specified time period. For example, oxidation controller **61** may activate oxidant supply **11** via supply control signal for an oxidation feed of oxidant **10** (e.g., atmospheric air) into oxidation mixer **51** at a 100% oxidation feed rate of oxidized gas **10** for thirty (30) seconds to one (1) minute (e.g., the feed rate in cubic feet/minute over a duration exceeding at least four (4) times a volume of oxidation chamber **52a** and retention chamber **53a**);
2. after completion of the purge, oxidation controller **61** controlling an activation of oxidant supply **11** via supply control signal SC for an oxidation feed of oxidant **10** into oxidation mixer **51** at a 50% oxidation feed rate; and
3. concurrent with or subsequent to the 50% oxidation feed rate of oxidant **10** into oxidation mixer **51**, oxidation

controller **61** executing a zero crossover regulation of heating element **55** via heating regulation signal HR.

Upon executing stage **S96**, oxidation controller **61** proceeds to a stage **S98** of flowchart **90** encompassing an oxidation chamber ignition check ("OCIC") by oxidation controller **61** involving a comparison by oxidation controller **61** of flash point temperature T_{FP} relative to a oxidation chamber ignition threshold ("OCIT") (e.g., 1100° F.).

Oxidation controller **61** cycles through stage **S98** until such time flash point temperature T_{FP} is equal to or greater than oxidation chamber ignition threshold OCIT whereby oxidation controller **61** proceeds to a stage **S100** of flowchart **90** to execute an oxidation stage of the thermal oxidation of waste gas stream **32** sequentially involving:

1. if spark ignitor **57** is employed, oxidation controller **61** controlling a spark ignition of combustible waste gas stream **70** within oxidation chamber **52a** by activating spark igniter **57** via a spark ignition signal **S1** provided to a power supply of spark igniter **57** (not shown for clarity)(e.g., a spark plug transformer based power circuit). The spark ignition is executed to facilitate the primary combustion reaction **71** of combustible waste gas stream **70** within oxidation chamber **52a**; and

2. oxidation controller **61** controlling an activation of a waste gas feed (e.g., 0.35 cubic feet/minute; 21 cubic feet/hour) into oxidation mixer **51** by opening control flow conduit **33** via valve control signal VC to form combustible waste gas stream **70** whereby oxidant **10** flowing into oxidation mixer **51** will create vacuum on waste gas stream **30** at the inlet of the oxidization mixer **51**; and

3. oxidation controller **61** controlling a regulation of a secondary combustion reaction of oxygenated waste gas stream **72** within retention chamber **53a** by modulating the oxidation feed rate of oxidant **10** by oxidant supply **11** based on a comparison of oxidation point temperature T_{OP} relative to a oxidation feed modulation threshold ("OFMT") (e.g., 1400° F.). For example, the oxidation feed rate of oxidant **10** is increased if the oxidation point temperature T_{OP} is less than the oxidation feed modulation threshold OFMT and conversely, the oxidation feed rate of oxidant **10** is decreased if the oxidation point temperature T_{OP} is greater than the oxidation feed modulation threshold OFMT.

Upon executing stage **S100**, oxidation controller **61** proceeds to a stage **S102** of flowchart **90** encompassing an oxidation protection check ("OPC") by oxidation controller **61** concurrently or sequentially involving:

1. a comparison by oxidation controller **61** of flash point temperature T_{FP} relative to a flash point protection threshold ("FPPT");

2. a comparison by oxidation controller **61** of oxidation point temperature T_{OP} relative to an oxidation point protection threshold ("OPPT"); and

3. a comparison by oxidation controller **61** of venting point temperature T_{VP} relative to a venting point protection threshold ("VPPT").

In practice, to ensure oxidation protection of thermal oxidizer **50**, the protection thresholds should represent a predefined maximum temperature of oxidation chamber **52a**, retention chamber **53a** and heat dissipater **54a** during oxidation stage **S100** (e.g., 1400° F. for oxidation chamber **52a**, 1800° F. for retention chamber **53a** and 300° F. for heat dissipater **54a**).

For a OPC cycle of stage **S102**, if flash point temperature T_{FP} is less than flash point protection threshold FPPT, AND if oxidation point temperature T_{OP} is less than oxidation point protection threshold OPPT, AND if venting point temperature T_{VP} is less than venting point protection thresh-

old VPPT, then oxidation controller **61** proceeds to a stage **S104** of flowchart **90** to be subsequently described herein.

Otherwise, if flash point temperature T_{FP} is equal to or greater than flash point protection threshold FPPT, OR if oxidation point temperature T_{OP} is equal to or greater than oxidation point protection threshold OPPT, OR if venting point temperature T_{VP} is equal to or greater than venting point protection threshold VPPT, then oxidation controller **61** proceeds to a stage **S106** of flowchart **90** to execute a shutdown stage of the thermal oxidation of waste gas stream **32** concurrently or sequentially involving:

1. oxidation controller **61** controlling a termination of waste gas feed into oxidation mixer **51** by closing control flow conduit **33** via valve control signal VC;

2. oxidation controller **61** controlling a deactivation of the zero crossover regulation of heating element **55** via heat regulation signal HR; and

3. oxidation controller **61** controlling an attenuation of the oxidation feed rate of oxidant **10** to zero (0) via supply control signal SC.

Upon completion of stage **S106**, oxidation controller **61** executes a power-off routine.

Still referring to FIG. **4C**, stage **S104** encompasses a waste gas limit check ("WGLC") by oxidization controller **61** involving:

1. a comparison by oxidation controller **61** of oxidation point temperature T_{OP} relative to a waste gas limit threshold ("WGLT"); and

2. a comparison by oxidation controller **61** of an open duration OD_{CVS2} of control flow conduit **33** relative to a fixed time period X in any unit of time (e.g., seconds as shown).

In practice, the waste gas limit threshold represents a predefined temperature indicative of a failure to achieve a primary combustion reaction **71** within oxidation chamber **53a** after time period X (e.g., ten (10) seconds) to thereby impede a flow of an unacceptable quantity of waste gas stream **32** through chambers **52a** and **53a**.

For a WGLP cycle of stage **S104**, if oxidation point temperature T_{OP} is equal to or greater than waste gas limit threshold WGLT OR if open duration OD_{CVS2} of control flow conduit **33** is less than X seconds, then oxidation controller **61** returns to stage **S102** as previously described herein.

Otherwise, if oxidation point temperature T_{OP} is less than waste gas limit threshold AND if open duration OD_{CVS2} of control flow conduit **33** is equal to or greater than X seconds, then oxidation controller **61** proceeds to stage **S106** as previously described herein.

Referring back to FIG. **3**, previously stated herein, upon oxidation controller **61** being powered-on, data logger **62** initiates a recording of flash point temperature T_{FP} , oxidation point temperature T_{OP} and venting point temperature T_{VP} over a course of an execution of flowchart **90** by oxidation controller **61**.

Furthermore, oxidation controller **61** may provide a local reporting of flash point temperature T_{FP} , oxidation point temperature T_{OP} and venting point temperature T_{VP} via a graphical display and/or data reporter **65** may provide a remote reporting of flash point temperature T_{FP} , oxidation point temperature T_{OP} and venting point temperature T_{VP} via a wireless communication to network **80**.

In practice, oxidation controller **61** and/or data logger **63** may be further structurally configured to detect conditions suitable for a warning (e.g., a visual and/or audial alarm, and/or a text message) to be locally issued by oxidation controller **61** or remotely issued by data reporter **65**. Such

11

conditions include, but are not limited to, (1) a continual cycling of stage S94 indicative of an operational issue with thermal oxidizer 50, (2) an inability of oxidation chamber 52a to reach oxidation combustion ignition threshold OCIT during stage S98, and (3) an occurrence offstage S104 as an indication of an operational issue of thermal oxidizer 50.

In practice, the structural dimensions and material compositions of oxidation chamber 52, retention chamber 53a and heat dissipater 54a will be dependent upon the specific oxidation application (e.g., 20,000 BTU/hour of waste gas oxidation).

To facilitate a further understanding of the inventions of the present disclosure, the following description of FIGS. 5-7 teach a couple of embodiments of the thermal oxidation system of FIG. 2. From this description; those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using numerous and various additional embodiments of thermal oxidization systems of the present disclosure.

Referring to FIGS. 5A and 5B, an embodiment 150a of thermal oxidizer 50a (FIG. 2) employs:

1. oxidant supply 11 (FIG. 2) in the form of an air blower 111a;
2. oxidation mixer 51 (FIG. 2) in the form of a venturi air-gas mixer 151a (not shown in FIGS. 5A and 5B; shown in FIG. 6A);
3. a control flow conduit 33 in the form of a solenoid valve 133a equipped with a flame arrestor 133b;
4. oxidation chamber 52a (FIG. 2) in the form an oxidation chamber 152 including a cylindrical insulation blanket and a cylindrical thermal mass defining an oxidation fluid path with a spiral heating element 155 disposed within the oxidation fluid path;
5. retention chamber 53a (FIG. 2) in the form of a retention chamber 153 including a cylindrical insulation blanket and a cylindrical thermal mass 154b defining a retention fluid path;
6. heat dissipater 54a (FIG. 2) in the form of a heat dissipater 154;
7. a flash thermocouple 67 (FIG. 2) in the form of a flash thermocouple 167;
8. an oxidization thermocouple 68 (FIG. 2) in the form of an oxidization thermocouple 168; and
9. a vent thermocouple 69 (FIG. 2) in the form of a vent thermocouple 169.

Also shown in FIGS. 5A and 5B is a control box 160 as an embodiment of control box 60 (FIG. 3).

FIG. 6A provides a view of venturi air-gas mixer 151a.

FIG. 6B provides a view of a nozzle 151b of venturi air-gas mixer 151a, an interior wall of oxidization chamber 153, a heating element 155 helically disposed within interior wall of oxidization chamber 153 and an interior wall of retention chamber 154.

FIG. 6C provides a view of pleated heat exchanger 157 of heat dissipater 154 encircling a top portion of the interior wall of retention chamber 154.

Referring to FIG. 7, an embodiment 150b of thermal oxidizer 50a (FIG. 2) is a modification of thermal oxidizer 150a (FIG. 5A) involving a substitution of air blower 11 with a flashback preventer 111b.

To facilitate a further understanding of inventions of the present disclosure, the following description of FIG. 8 further teaches basic inventive principles of a thermal oxidization system and a thermal oxidization method of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and

12

using numerous and various embodiments of thermal oxidization systems and thermal oxidization methods of the present disclosure.

Referring to FIG. 8, a thermal oxidization system of the present disclosure incorporates a thermal oxidizer 50b employing oxidation mixer 51, an oxidation chamber 52a, a retention chamber 53b and a heat dissipater 54b forming a fluid flow path for an oxidation of a waste gas stream 32 (e.g., TRI gases) into oxidized combustion products 74 (e.g., oxidized gases 73 (FIG. 2) including CO₂, H₂O, N₂ and O₂).

As previously described, oxidation mixer 51 is structurally configured for mixing an oxidant 10 supplied by oxidant supply 11 (e.g., a blower or a pump) via a supply line 12 and a waste gas stream 32 (e.g., TRI gases) supplied by waste gas source 30 as controlled via a control flow conduit 33 (e.g., a solenoid valve and a flame arrestor in series coupling waste gas source 30 to oxidation mixer 51) into a combustible waste gas stream 70.

In one embodiment, oxidation mixer 51 is a venturi air-gas mixer whereby turbulent fluid flows of oxidant 10 and waste gas stream 32 into the venturi air-gas mixer are controlled via oxidant supply 11 and control flow conduit 33 to ensure combustible waste gas stream 70 attains proportional concentrations of oxidant 10 and waste gas stream 32 within a flammable range (e.g., 11.5:1 oxidant 10 to waste gas stream 32). Additionally, oxidation mixer 51 may be equipped with a nozzle (not shown) for regulating a feeding of combustible waste gas stream 70 into oxidation chamber 52b whereby the nozzle may be structurally configured to generate more turbulence to combustible waste gas stream 70.

Oxidation chamber 52a is structurally configured for implementing a primary combustion reaction 71 therein of combustible waste gas stream 70 via a controlled emission of heat waves 56a by a heating element 55a. In one embodiment, oxidation chamber 52a is a refractory ceramic cylinder and heating element 55a is embedded within the walls of the refractory ceramic cylinder.

Optionally, oxidation chamber 52a may further employ a spark igniter 66 for a controlled ignition of combustible waste gas stream 70 at a proximal opening of oxidation chamber 52a. For this embodiment, if oxidization mixer 51 is equipped with a nozzle, then a distal tip of spark igniter 66 may be positioned within or adjacent to the flow of the combustible waste gas stream 70 out of the nozzle into oxidation chamber 52a.

Combustible waste gas stream 70a may fully or partially combust within oxidation chamber 52a. Oxidation chamber 52a therefore includes a supplement air inlet 52b for an additional flow of oxidant 10 via oxidant supply line 14 or via atmosphere to supplement the oxygenated waste gas stream 72 flowing into retention chamber 53b.

Retention chamber 53b is structurally configured for implementing a secondary combustion reaction 76 therein oxygenated waste gas stream 72 via a controlled omission of heat waves 56b by a heating element 55b. In one embodiment, retention chamber 53b is a refractory ceramic cylinder and heating element 55b is embedded within the walls of the refractory ceramic cylinder integrated with oxidation chamber 53b as shown. Retention chamber 53b ensures a complete oxidation of oxygenated waste gas stream 72 into heated oxidized combustion products 74 (e.g., oxidized gases 73 (FIG. 2) including CO₂, H₂O, N₂ and O₂).

Heat dissipater 54b is structurally configured for implementing a heat exchange with heated oxidized combustion products 74 to vent cooled oxidized combustion products 74 into the atmosphere. In one embodiment, heat dissipater 54b

13

includes a nozzle **59** having a cylindrical shape whereby oxidized combustion products **74** exits heat dissipater **54b** along a length and circumference of heat exchanger **57** with cooling atmosphere air **75** being directed vertically past heat exchanger **57** to thereby extract heat from heat exchanger **57**.

The thermal oxidation system of FIG. **8** further incorporates a control system employing an oxidation controller **61**, a data logger **63** (FIG. **3**) and a data reporter **65** (FIG. **3**) housed within a control box **60** (FIG. **3**).

Oxidation controller **61** is structurally configured for controlling an operation of thermal oxidizer **50b** in accordance with the thermal oxidation method of FIGS. **4A-4C** as previously described herein. More particularly, heating elements **55a** and **55b** are controlled via a similar scheme as heating element **55** (FIG. **2**).

In practice, oxidation controller **61** may be omitted whereby the heating elements **55** of thermal oxidizer **50a** (FIG. **2**) and thermal oxidizer **50b** (FIG. **8**) may be operated via an on/off switch.

Also in practice, oxidation controller **61** may implement a variation of the thermal oxidation method **90** (FIGS. **4A-4C**) excluding the heat dissipater thermocouple **69** and involving the oxidation chamber thermocouple **67** and/or the retention chamber thermocouple **68**. For such an embodiment, a one loop or a two loop proportional-integral-derivative ("PID") control may be implemented by oxidation controller **61** involving a zero crossover regulation of the heating element(s) **55** via heating regulation signal HR.

To facilitate a further understanding of the inventions of the present disclosure, the following description of FIGS. **9-10** teach a couple of embodiments of the thermal oxidation system of FIG. **8**. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using numerous and various additional embodiments of thermal oxidation systems of the present disclosure.

Referring to FIG. **9**, an embodiment **250** of thermal oxidizer **50b** (FIG. **8**) employs:

1. an oxidation mixer nozzle **251** having an air inlet on a side wall and a bottom gas inlet;
2. oxidation chamber **52a** (FIG. **8**) in the form an oxidation chamber **252** including a cylindrical insulation blanket and a cylindrical thermal mass defining an oxidation fluid path with a spiral heating element **255a** encircling the oxidation fluid path;
3. retention chamber **53b** (FIG. **8**) in the form of a retention chamber **253** including an insulation blanket and a thermal mass defining a retention fluid path with a spiral heating element **255b** encircling the oxidation fluid path;
4. heat dissipater **54b** (FIG. **10A**) in the form of a heat dissipater **254** having a rain cap **290** (FIG. **9**); and
5. mounting brackets **291a** and **291b**.

Thermocouples may or may not be incorporated dependent upon the control scheme.

For example, FIG. **10A** illustrates an atmospheric version **250a** of thermal oxidizer **250** (FIG. **9**) involving a non-force flow of atmospheric air into the thermal oxidizer **250a** and an on/off control of the heating elements. A heat exchanger **257** provides for heat exchange and back flash protection of the heat dissipater, and a back flash **292** protects a back flash of the oxidation mixer.

By further example, FIG. **10B** illustrates a force-air version **250b** of thermal oxidizer **250** (FIG. **9**) incorporating an air blower **211**, a solenoid valve **233a**, a flash arrestor **233b**, an oxidation thermocouple **267** and a retention thermocouple **268**.

14

Referring to FIGS. **1-10**, those having ordinary skill in the art of the present disclosure will appreciate various advantages of the inventions of the present disclosure including, but not limited to:

1. an active or passive control of an efficient thermal combustion of toxic release inventories gases ("TRI"), particularly in compliance with applicable environment regulations; and
2. data logging and data reporting of the active control of the thermal combustion of TRI.

To further facilitate an understanding of inventions of the present disclosure, the following description of FIGS. **11-14** teach basic inventive principles of a gas fired based thermal oxidation systems and gas fired based thermal oxidation methods of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of gas fired based thermal oxidation systems and gas fired based thermal oxidation methods of the present disclosure.

Referring to FIG. **11**, an exemplary embodiment of a gas fired thermal oxidation system of the present disclosure incorporates a thermal oxidizer **300** employing an oxidation mixer **310** in the form of a venturi air-gas mixer, an oxidation chamber **302**, a retention chamber **303** and a heat dissipater (not shown) forming a fluid flow path for oxidation of a waste gas stream (e.g., TRI gases) into oxidized combustion products (e.g., oxidized gases including CO₂, H₂O, N₂ and O₂). As will be subsequently described in more detail herein, a gas fire based thermal oxidation method of the present disclosure represented by a flowchart **400** (FIG. **12**) implements an operation of thermal oxidizer **300** involving an ignition phase as best shown in FIG. **13A** for heating up chambers **302**, **303** and a combustion phase as best shown in FIG. **13B** for generating oxidized combustion products.

Oxidation mixer **310** is structurally configured for implementing an ignitable mixture of an oxidant **10** and natural gas **35** during the ignition phase of the thermal oxidation method. The oxidant **10** is supplied to oxidation mixer **310** by an oxidant supply (not shown) (e.g., a blower or a pump) as controlled via a control flow conduit **312** (e.g., a solenoid and a modulation valve in series coupling the oxidant supply to oxidation mixer **310**). The natural gas **35** is supplied to oxidation mixer **310** by a natural gas source and a flame arrestor (not shown) as controlled via a control flow conduit **314** (e.g., a solenoid valve and a modulation valve in series coupling the natural gas source to oxidation mixer **310**).

Oxidation mixer **310** is further structurally configured for implementing a combustible mixture of oxidant **10** and waste gas **32** during the combustion phase of the thermal oxidation method. Again, oxidant **10** is supplied to oxidation mixer **310** by an oxidant supply (not shown) (e.g., a blower or a pump) as controlled via a control flow conduit **312** (e.g., a solenoid valve and a flame arrestor in series coupling the oxidant supply to oxidation mixer **310**). The waste gas **32** is supplied to oxidation mixer **310** by a waste gas source and a flame arrestor (not shown) as controlled via a control flow conduit **315** (e.g., a solenoid valve and a modulation valve in series coupling the waste gas source to oxidation mixer **310**).

In one exemplary embodiment, oxidation mixer **310** is a venturi air-gas mixer whereby turbulent fluid flows of oxidant **10** and waste gas **32** into the venture air-gas mixer are controlled via control flow conduits **312** and **316** to ensure

15

proportional concentrations of oxidant **10** and waste gas stream **32** within a flammable range (e.g., **11.5:1** oxidant **10** to waste gas stream **32**).

Still referring to FIG. **11**, oxidation mixer **310** is equipped with a burner nozzle **311** having a mounted spark igniter for igniting the ignitable mixture of oxidant **10** and natural gas **35** via oxidation mixer **310** during the ignition phase of the thermal oxidation method.

During the combustion phase of the thermal oxidation method, burner nozzle **311** is further structurally configured for generating turbulence to the combustible mixture of oxidant **10** and waste gas **32** (i.e., a combustible waste gas stream) via oxidation mixer **310** and for regulating a feeding of the combustible mixture of oxidant **10** and waste gas **32** into oxidation chamber **302**.

Still referring to FIG. **11**, during the ignition phase of the thermal oxidation method, oxidation chamber **302** and retention chamber **303** are structurally configured to absorb a flame generated by oxidation mixer **310** and burner nozzle **311** for elevating a thermal mass temperature within the chambers **302**, **303** above a thermal mass combustion threshold (e.g., $>1450^{\circ}\text{F}$).

During the combustion phase of the thermal oxidation method, oxidation chamber **302** is further structurally configured for implementing a primary combustion reaction **71a** therein of the combustible waste gas stream **10**, **32** into an oxygenated waste gas stream, and retention chamber **303** is structurally configured for implementing a retention time for a secondary combustion reaction **71b** of the oxygenated waste gas stream into heated oxidized combustion products (e.g., oxidized gases including CO_2 , H_2O , N_2 and O_2).

In one exemplary embodiment, oxidation chamber **302** and retention chamber **303** are integrally formed as a refractory ceramic cylinder with a diffuser **301** separating the chambers **302**, **303**.

Still referring to FIG. **11**, the heat dissipator (not shown) may be structurally configured for implementing a heat exchange of atmosphere air with heated oxidized combustion products to vent cooled oxidized combustion products into the atmosphere. Alternatively, the heat dissipator may be structurally configured for implementing a vaporizing of oxidized combustion products **74** for a subsequent venting of some of the products to atmosphere (e.g., oxidized gases N_2 and O_2) and for a subsequent capture and storage of other products (e.g., oxidized gas CO_2).

Still referring to FIG. **11**, the thermal oxidation system of the present disclosure further incorporates a control system employing an oxidation controller **360**, a data logger (not shown) and a data reporter (not shown).

Oxidation controller **360** is structurally configured for controlling an operation of thermal oxidizer **300** as will be further described herein in connection with a description of flowchart **400** of FIG. **12**.

In one exemplary embodiment, oxidation controller **360** is an application specific main board or an application specific integrated circuit for controlling a thermal oxidation application of various inventive principles of the present disclosure as subsequently described herein in connection with FIG. **12**. The structural configuration of oxidation controller **360** may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for controlling an execution of the thermal oxidation application.

16

A non-limiting example of oxidation controller **360** is an all-in-one XL4 controller sold by Homer APG of Indianapolis, Indiana that is configured in accordance with the inventive principles of the present disclosure.

The data logger is structurally configured for logging operational data transmitted by oxidation controller **360** to the data logger via a push or pull operation, or by a monitoring of specific data points of oxidation controller **360** by the data logger. The operational data includes data informative of an operational status of thermal oxidizer **300** in executing the oxidation of waste gas stream **32**.

In one exemplary embodiment, the data logger is an application specific main board or an application specific integrated circuit for controlling a data logging application of the present disclosure. The structural configuration of the data logger may include, but is not limited to, processor(s), computer-usable/computer readable storage medium(s), an operating system, application module(s), peripheral device controller(s), slot(s) and port(s). Each application module consists of an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for executing the data logging application.

A non-limiting example of the data logger is an all-in-one XLE controller sold by Homer APG of Indianapolis, Indiana that is configured in accordance with the inventive principles of the present disclosure. Another non-limiting example of the data logger is as an application module configured within oxidation controller **360**.

Also in practice, the data logger may be omitted and oxidation controller **360** may be configured for executing the data logging application of the present disclosure.

The data reporter **65** is structurally configured for receiving reporting data from the data logger via a push or pull operation and transmitting the reporting data to a network (e.g., a cellular network). The reporting data is the operational data processed by the data logger into a reporting format associated with the data reporter.

In one embodiment, the data reporter is an application specific modem for executing a data reporting application of the present disclosure.

A non-limiting example of the data logger **66** is a SonicU™ modem sold by SonicU, LLC of Greenfield, Indiana that is configured in accordance with the inventive principles of the present disclosure.

Referring back to FIG. **11**, oxidation controller **360** regulates a control of thermal oxidizer **300** in accordance with the thermal oxidation method of the present disclosure as represented by a flowchart **400** shown in FIG. **12**. In support of executing the thermal oxidation method, (a) oxidation controller **360** senses and controls the generation of the flame via a connection of a flame detector **304** within combustion chamber **302** at a signal input **361s** of oxidation controller **360**, (b) oxidation controller **360** senses and controls a thermal mass temperature of retention chamber **303** via a connection of a thermocouple **305** within retention chamber **303** at a signal input **362s**, (c) oxidation controller **360** controls an activation and a deactivation of the spark igniter of burner nozzle **311** via a signal output **363s**, (d) oxidation controller **360** controls a feeding rate of natural gas **35** into oxidation mixer **310** via a connection of a signal output **364s** to control flow conduit **312**, (e) oxidation controller **360** controls a feeding rate of waste gas **32** into oxidation mixer **310** via a connection of a signal output **365s** to control flow conduit **314**, and (f) oxidation controller **360**

17

controls a feeding rate of oxidant **10** into oxidation mixer **310** via a connection of a signal output **366s** to control flow conduit **316**.

Referring to FIGS. **11** and **12**, prior to be oxidation controller **360** being powered-on, control flow conduits **312**, **314**, **316** are closed. Upon oxidation controller **360** being powered-on, oxidation controller **360** initiates a thermocouple validity check of thermocouple **305** involving a comparison by oxidation controller **360** of a thermal mass temperature of retention chamber **303** as sensed by thermocouple **305** to a thermocouple validity threshold. In practice, to test the validity of the thermocouple **305**, the thermocouple validity threshold should be set at an anticipated maximum room temperature (e.g., 22° C.) or to an anticipated maximum local outdoor air temperature.

For the thermocouple validity test, if the thermal mass temperature of retention chamber **303** is more than the thermocouple validity threshold, then oxidation controller **360** is powered down for a maintenance inspection of thermocouple **305** and powered on again after a successful repair/replacement, if any, of the thermocouple **305**.

Otherwise, if the thermal mass temperature of retention chamber **303** is equal to or less than the thermocouple validity threshold, then oxidation controller **360** initiates a thermal oxidation limit check involving a comparison by oxidation controller **360** of a thermal mass temperature of retention chamber **303** as sensed by thermocouple **305** to a thermal oxidation limit threshold.

In practice, to ensure a safe heat activation of chambers **302,303**, the thermal oxidation limit threshold should represent a predefined maximum temperature of retention chamber **303** prior to the heat activation of thermal oxidizer **300** (e.g., 300° C.) for all thermocouple limit thresholds).

For the thermal oxidation limit check, if the thermal mass temperature of retention chamber **303** is more than the thermal oxidation limit threshold, then oxidation controller **360** is powered down for a cooling period whereby oxidation controller **360** may be powered up again upon expiration of the cooling period to re-execute the thermal oxidation limit check.

Otherwise, if the thermal mass temperature of retention chamber **303** is equal to or less than the thermal oxidation limit threshold, then oxidation controller **360** executes a stage **S402** of flowchart **400** encompassing a natural gas flame ignition involving (a) an activation of the spark igniter of burner nozzle **311** via signal output **363s**, (b) a controlled feeding rate of natural gas **35** into oxidation mixer **310** via signal output **364s**, and (c) a controlled feeding rate of oxidant **10** into oxidation mixer **310** via signal output **366s**.

Referring to FIGS. **12** and **13A**, during stage **S402**, to reach a thermal combustion threshold (e.g., 1450° F.), oxidation controller **360** may implement a modulation of the natural gas flame ignition involving (a) an increase in the feeding rate of natural gas **35** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** decreases, (b) a decrease in the feeding rate of natural gas **35** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** increases, (c) an increase in the feeding rate of oxidant **10** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** increases and (d) a decrease in the feeding rate of oxidant **10** into oxidation mixer **310** as the thermal mass temperature of retention chamber **303** decreases.

In practice of stage **S402**, any increase or decrease of the feeding rate of natural gas **35** into oxidation chamber **310** is inversely proportional to any increase or decrease of the thermal mass temperature of retention chamber **303**, and any

18

increase or decrease of the feeding rate of natural gas **35** into oxidation chamber **310** is directly proportional to any increase or decrease of the thermal mass temperature of retention chamber **303**.

Upon the thermal mass temperature of retention chamber **303** being more than a thermal mass combustion threshold as determined by oxidation controller **360** during a stage **S404** of flowchart **400**, oxidation controller **360** executes a stage **S406** of flowchart **400** encompassing a waste gas combustion involving (a) a deactivation of the spark igniter of burner nozzle **311** via signal output **363s**, (b) a reduced feeding rate of natural gas **35** into oxidation mixer **310** via signal output **364s** or a closing of control flow conduit **312** via signal output **364s**, (c) a controlled feeding rate of waste gas **32** into oxidation mixer **310** via signal output **365s**, and (d) a controlled feeding rate of oxidant **10** into oxidation mixer **310** via signal output **366s**.

Referring to FIGS. **12** and **13B**, during stage **S406**, to maintain a constant thermal mass temperature of retention chamber **303** (e.g., 1450° F.), oxidation controller **360** may implement a modulation of the waste gas combustion involving (a) a closing of control flow conduit **314** into oxidation mixer **310** via output signal **365s** when the thermal mass temperature of retention chamber **303** is more than the constant thermal mass temperature of retention chamber **303** and (b) an increase in the feeding rate of oxidant **10** into oxidation mixer **310** via output signal **366s** when the thermal mass temperature of retention chamber **303** is more than the constant thermal mass temperature of retention chamber **303**.

Also during stage **S406**, if the thermal mass temperature of retention chamber **303** becomes more than the thermal oxidation limit threshold as determined by oxidation controller **360** during a stage **S408** of flowchart **400**, the oxidation controller **360** executes a stage **S410** of flowchart **400** encompassing an oxidizer shut down involving (a) a continued deactivation of the spark igniter of burner nozzle **311** via signal output **363s**, (b) a closing of control flow conduit **312** into oxidation mixer **310** via output signal **364s**, (c) a closing of control flow conduit **314** into oxidation mixer **310** via output signal **365s**, and (d) a closing of control flow conduit **316** into oxidation mixer **310** via output signal **366s**.

Otherwise, oxidation controller **360** will continue to execute stages **S402-S408** as a function of the thermal mass temperature of retention chamber **303** as previously described herein for stages **S402-S408**.

Referring to FIG. **14**, a thermal oxidizer **300'** is a version of thermal oxidizer **300** (FIG. **11**) that includes a supplement air inlet **306** for an additional flow of oxidant **10** via control flow conduit **316** as shown or via atmosphere to supplement the oxygenated waste gas stream flowing into retention chamber **303**.

To further facilitate an understanding of inventions of the present disclosure, the following description of FIGS. **15** and **16** teaches basic inventive principles of a greenhouse gas capture based thermal oxidation system of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of greenhouse gas capture based thermal oxidation systems of the present disclosure.

Referring to FIG. **15**, an exemplary greenhouse gas capture based thermal oxidation system of the present disclosure employs an embodiment **500** of a thermal oxidizer as previously described in the present disclosure, such as, for

19

example, thermal oxidizer **50a** of FIG. 2, thermal oxidizer **50b** of FIG. 8, thermal oxidizer **300a** of FIG. 11 and/or thermal oxidizer **300b** of FIG. 14.

In operation, in accordance with the inventive principles of the present disclosure as set forth herein, thermal oxidizer **500** mixes an oxidant (e.g., atmospheric air) and waste gas (e.g., TRI gases) to yield a combustible waste gas stream **501** that sequentially undergoes a primary combustion reaction **502** into an oxygenated waste gas stream (not shown), a secondary combustion reaction **503** of the oxygenated waste gas stream into oxidized combustion products **504** (e.g., oxidized gases including CO₂, H₂O, N₂ and O₂ as shown). A final operational phase of thermal oxidizer involves a heat dissipation of oxidized combustion products **504** (e.g., a heat dissipation of >1750° F. to ≈300° F.).

Still referring to FIG. 15, the exemplary greenhouse gas processor based thermal oxidation system of the present disclosure further employs a greenhouse gas processor **700** in fluid communication with thermal oxidizer **500** (e.g., via a pipeline **600** as shown) to receive oxidized combustion products **504** at the reduced temperature. In one exemplary embodiment, thermal oxidizer **500** may include a heat exchanger to vent the oxidized combustion products **504** at the reduced temperature with the heat exchanger being enclosed by a sealed duct connector or the like to enable an unregulated or a regulated flow of the oxidized combustion products **504** at the reduced temperature to greenhouse gas processor **700**. In a second exemplary embodiment, thermal oxidizer **500** may include a vaporizer integrated with a pipeline or the like to enable an unregulated or a regulated flow of a water vaporization of the oxidized combustion products **504** to greenhouse gas processor **700** as will be further described herein.

For purposes of describing and claiming the inventions of the present disclosure, greenhouse gas processor **700** broadly encompasses any machine in accordance with embodiments described in the present disclosure, as known in the art of the present disclosure and hereinafter conceived operable to extract greenhouse gas(es) **701** (e.g., oxidized gas CO₂ as shown) of oxidized combustion products **504** from non-greenhouse gas(es) **702** (e.g., oxidized gases N₂ and O₂ as shown). Thereafter, greenhouse gas processor **700** may vent non-greenhouse gas(es) **702** to atmosphere via a pipeline **601**, and may control a transportation and/or a storage of greenhouse gas(es) **701** via a pipeline **602**.

In one exemplary embodiment, greenhouse gas processor **700** may water saturate the oxidized combustion products into a liquid condensate to extract the greenhouse gas(es) **701** within oxidized combustion products **504** from non-greenhouse gas(es) **702** via a greenhouse gas condensation process in accordance with embodiments of the present disclosure, as known in the art of the present disclosure or hereinafter conceived. For example, the greenhouse gas condensation process may involve a fog spraying of water onto the water vaporization of oxidized combustion products **504** to yield a liquid condensation of the greenhouse gas(es) **702** followed by an acid neutralization of the liquid condensation of the greenhouse gas(es) **702** as will be further described herein.

Referring to FIG. 16, an exemplary embodiment of the greenhouse gas capture based thermal oxidation system of FIG. 15 employs an embodiment **500a** of the thermal oxidizer including an oxidation mixer **510**, an oxidation chamber **520**, a retention chamber **530**, and a heat dissipator in the form of a vaporizer **540**. In practice, vaporizer **540** broadly encompasses any device in accordance with embodiments of the present disclosure, as known in the art

20

of the present disclosure and hereinafter conceived operable to utilizing a vaporizing agent (e.g., water) to facilitate a vaporization **504a** of the oxidized combustion products **504** flowing from retention chamber **530** to vaporizer **540** to thereby effect a heat dissipation of oxidized combustion products **504** (e.g., a reduction of heated oxidized gases from 1750° F.-1800° F. to ≈300° F.).

In operation, in accordance with the inventive principles of the present disclosure as set forth herein, oxidation mixer **510** mixes an oxidant (e.g., atmospheric air) and waste gas (e.g., TRI gases) to yield a combustible waste gas stream (not shown), a primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream (not shown) occurs in oxidation chamber **520**, and a secondary combustion reaction of the oxygenated waste gas stream into heated oxidized combustion products occurs in retention chamber **530**.

For this embodiment, vaporizer **540** is in fluid communication with a water supply **560** via a water isolation valve **603** and a water control valve **604** to fog spray of water onto the oxidized combustion products of retention chamber **530** flowing into vaporizer **540** to yield a water vaporization **504a** of oxidized combustion products **504** consisting of water vapor mixed with oxidized combustion products **504** at a reduced temperature. For example, a temperature of oxidized combustion products **504** exiting retention chamber **530** into vaporizer **540** may be approximately 1750° F. and vaporizer **540** may cool the oxidized combustion products **540** via the vaporization to a temperature of 300° F.

Still referring to FIG. 16, the exemplary embodiment of the greenhouse gas capture based thermal oxidation system of FIG. 15 further employs an embodiment **700a** of the greenhouse gas processor **700** including a condenser **710** and an acid neutralizer **720**.

In practice, condenser **710** broadly encompasses any device in accordance with the present disclosure, as known in the art of the present disclosure or hereinafter conceived, that is operable to water saturate the water vaporization of the oxidized combustion products to a dew point temperature, and an acid neutralizer **720** broadly encompasses any device in accordance with the present disclosure, as known in the art of the present disclosure or hereinafter conceived for neutralizing acidic liquid.

In operation, condenser **710** receives the water vaporization **504a** of oxidized combustion products **504** and is in fluid communication with water supply **560** via a water isolation valve **605** and a water control valve **606** to fog spray water onto the water vaporization **504a** of oxidized combustion products **504** to yield a saturation **504b** of oxidized combustion products **504** at a further reduced temperature. For example, condenser **710** may water saturate **504b** the vaporization **504a** of oxidized combustion products **504** to below a dew point temperature of 80° F. to yield an acidic greenhouse condensate (e.g., carbonic acid H₂CO₃ droplets as shown) and non-greenhouse gas vapors (e.g., N₂ and O₂ as shown).

Acid neutralizer **720** contains a neutralizing agent (e.g., a granular organic neutralizing agent like limestone, not shown) to facilitate neutralizing an acidic greenhouse condensate **701a** (e.g., neutral carbonic acid H₂CO₃). Acid neutralizer **720** thereafter separates the non-greenhouse gas(es) **702a** (e.g., oxidized gases N₂ and O₂ as shown) from neutralized acidic greenhouse condensate **701a**, whereby the non-greenhouse gas(es) **702a** may be vented to atmosphere via pipeline **601** and a transportation and/or a storage of substantially neutralized acidic greenhouse condensate **701a** may be controlled via pipeline **602**.

21

To further facilitate an understanding of inventions of the present disclosure, the following description of FIGS. 17-19 teaches basic inventive principles of a greenhouse gas capture based thermal oxidization method of the present disclosure. From this description, those having ordinary skill in the art will appreciate how to apply the inventive principles of the present disclosure for making and using additional numerous and various embodiments of greenhouse gas capture based thermal oxidization methods of the present disclosure.

Referring to FIG. 17, a flowchart 800 represents an exemplary embodiment of a greenhouse gas capture based thermal oxidization method of the present disclosure. The following description of flowchart 800 will be in the context of a heat dissipating vaporizer 540a as shown in FIG. 18, a condenser 710a as shown in FIG. 19, and an acid neutralizer 720a as shown in FIG. 19.

Referring to FIG. 18, heat dissipating vaporizer 540a employs tray type vaporizing chamber 541 having a breather cover 543, a nozzle 542, a pressure sensor 544 embedded into vaporizing chamber 541, and a temperature sensor 546 embedded into vaporizing chamber 541.

Referring to FIG. 19, condenser 710a includes a condensing chamber 711, a nozzle 712, a vapor saturation chamber 713 and a temperature sensor 714 embedded into vapor saturation chamber 713. Acid neutralizer 720a includes a primary neutralizing chamber 722 enclosing a granular organic neutralizing material (e.g., limestone), a secondary neutralizing chamber 723 enclosing a granular organic neutralizing material (e.g., limestone), and a neutralized water chamber 724.

Referring to FIGS. 17 and 18, a stage S802 of flowchart 800 encompasses a water vaporization 504a of oxidized combustion products 504 (e.g., oxidized gases including CO₂, H₂O, N₂ and O₂) exiting retention chamber 530 into vaporizing chamber 541. More particularly, water supply 560 is controlled to supply a specified flow of water at a specified pressure to water control valve 604 (e.g., <1.0 gallons per hour at 60 psi). Temperature sensor 546 senses a temperature of vaporizing chamber 541 and a temperature controller 547 (e.g., a PID controller) controls water flow control valve 604 as a function of the sensed temperature of vaporizing chamber 541 to maintain a specified vaporization temperature point within vaporizing chamber 541 (e.g., 300° F.) as known in the art of the present disclosure. Water flowing through water flow control valve 604 is fogged sprayed via nozzle 542 onto the oxidized combustion products 504 flowing through the trays of vaporizing chamber 541 to yield a flow of the water vaporization 504a of oxidized combustion products 504 at a reduced temperature through pipeline 600. Pressure sensor 544 senses a pressure within vaporizing chamber 541 for purposes of controlling a flow rate of the water vaporization 504a of oxidized combustion products 504 through pipeline 600 as will be subsequently described herein.

Referring to FIGS. 17 and 19, a stage S804 of flowchart 800 encompasses a water saturation of vaporized oxidized combustion products 504a flowing into condenser 710a. More particularly, a motorized air pump 612 is operated via a motor speed controller 611 to pull the water vaporization 504a of oxidized combustion products 504 through pipeline 600 into condensing chamber 711. Motor speed controller 611 is controlled via a pressure controller 545 (e.g., a PID controller) to control the flow rate of the water vaporization 504a of the oxidized combustion products 504 through pipeline 600 into condensing chamber 711 as a function of

22

a pressure within vaporizing chamber 541 (FIG. 18) as sensed by pressure sensor 544 as known in the art of the present disclosure.

Water supply 560 is controlled to supply a specified flow of water at a specified pressure to water control valve 606 (e.g., <1.0 gallons per hour at 60 psi), and water flowing through water flow control valve 606 is fogged sprayed via nozzle 712 upon vaporized oxidized combustion products 504a flowing into fogging condensing chamber 711 to yield a water saturation of vaporized oxidized combustion products 504a in vapor saturation chamber 713. Temperature sensor 714 senses a temperature of vapor saturation chamber 713 and a temperature controller 715 (e.g., a PID controller) controls water flow control valve 606 as a function of the sensed temperature of vapor saturation chamber 713 to maintain a specified temperature point within condensation chamber 713 (e.g., 80° F.) as known in the art of the present disclosure. As a result, a condensation of greenhouse gas(es) drops through chamber 713 to the primary neutralizing chamber 722 of acid neutralizer 720a.

Still referring to FIGS. 17 and 19, a stage S806 of flowchart 800 encompasses an acid neutralization of the saturated oxidized combustion products. More particularly, the saturated oxidized combustion products downwardly percolate through the granular organic neutralizing material (e.g., limestone) of primary neutralizing chamber 722 whereby a substantially portion of the acidic greenhouse gas of the saturated oxidized combustion products dissolves into neutralized liquid 701a maintained in a neutralized water chamber 724, and the non-greenhouse gases of the saturated oxidized combustion products upwardly flowing through the granular organic neutralizing material (e.g., limestone) of secondary neutralizing chamber 723. The neutralized liquid 701a within neutralized water chamber 724 is continually or intermittently circulated via circulating pump 731 and a pipeline 732 to a nozzle 733 spraying the neutralized liquid 701a into secondary neutralizing chamber 723, whereby any residue of acidic greenhouse gas upwardly flowing through secondary neutralizing chamber 723 may also be dissolved into neutralized liquid 701a maintained in neutralized water chamber 724.

Still referring to FIGS. 17 and 19, a stage S808 of flowchart 800 encompassing a venting of the non-greenhouse gases into atmosphere and a stage S810 of flowchart 800 encompasses a storage of neutralized liquid 701a for future transportation. More particularly, the motorized air pump 612 will pull the non-greenhouse gas(es) from secondary neutralizing chamber 723 through pipeline 602 and an exhaust 613 to atmosphere. Additionally, a control float switch 740 within a neutralized water chamber 724 is operated to control a water flow control valve 742 to drain neutralized liquid 701a to a surge neutralized water tank 725 whenever a volume of neutralized liquid 701a within neutralized water chamber 724 exceeds a specified volume. Additionally, a pressure equalizing valve 751 and a pressure isolation valve are operated to relieve pressure within neutralized water chamber 724 as a function of control float switch 740.

While various embodiments of the present invention have been illustrated and described, it will be understood by those skilled in the art that the methods and the system as described herein are illustrative, and various changes and modifications may be made and equivalents may be substituted for elements thereof without departing from the true scope of the present invention. In addition, many modifications may be made to adapt the teachings of the present invention without departing from its central scope. There-

23

fore, it is intended that the present invention not be limited to the particular embodiments disclosed as the best mode contemplated for carrying out the present invention, but that the present invention include all embodiments falling within the scope of the appended claims.

The invention claimed is:

1. A thermal oxidization system, comprising:
 - a thermal oxidizer including an oxidation mixer, an oxidation chamber, a retention chamber and a heat dissipater forming a fluid flow path for a thermal oxidation of a waste gas,
 - wherein the oxidation mixer is structurally configured to facilitate a combustible mixture of a waste gas stream and an oxidant within the oxidation mixer into a combustible waste gas stream flowing within the oxidation mixer,
 - wherein the oxidation chamber is in fluid communication with the oxidation mixer to receive a flow of the combustible waste gas stream,
 - wherein the oxidation chamber includes a primary heating element or a gas burner to facilitate a primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream within the oxidation chamber,
 - wherein the retention chamber is in fluid communication with the oxidation chamber to receive a flow of the oxygenated waste gas stream,
 - wherein the retention chamber is structurally configured to facilitate a secondary combustion reaction of the oxygenated waste gas stream into oxidized gases within the retention chamber,
 - wherein the oxidized gases includes at least one greenhouse gas,
 - wherein the heat dissipater is in fluid communication with the retention chamber to receive a flow of the oxidized gases,
 - wherein the heat dissipater is structurally configured to heat dissipate a flow of the oxidized gases within the heat dissipator; and
 - a greenhouse gas processor in fluid communication with the heat dissipator to receive the oxidized gases flowing within the heat dissipator, wherein the greenhouse gas processor is structurally configured to extract the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor.
2. The thermal oxidization system of claim 1, wherein the heat dissipator structurally configured to heat dissipate the oxidized gases flowing within the heat dissipator includes:
 - the heat dissipator being structurally configured to facilitate a water vaporization of the oxidized gases flowing within the heat dissipator.
3. The thermal oxidization system of claim 2, further comprising:
 - a vaporization controller structurally configured to regulate the water vaporization by the heat dissipator of the oxidized gases flowing within the heat dissipator based on a temperature of the oxidized gases flowing within the heat dissipator.
4. The thermal oxidization system of claim 1, wherein the greenhouse gas processor structurally configured to extract the at least one greenhouse gas from the oxidized gases within the greenhouse processor includes:
 - the greenhouse gas processor being structurally configured to form an acidic greenhouse gas condensate within the greenhouse gas processor.
5. The thermal oxidization system of claim 4, wherein the greenhouse gas processor structurally configured to liquid

24

condensate the at least one greenhouse gas within the greenhouse gas processor includes:

- the greenhouse gas processor being structurally configured to facilitate a water saturation of the oxidized gases within the greenhouse gas processor.
6. The thermal oxidization system of claim 5, further comprising:
 - a condensation controller structurally configured to regulate the water saturation by the condenser of the oxidized gases flowing within the condenser based on a temperature of the oxidized gases flowing within the condenser.
 7. The thermal oxidization system of claim 1, wherein the greenhouse gas processor is further structurally configured to capture the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor.
 8. The thermal oxidization system of claim 1, wherein the greenhouse gas processor is further structurally configured to form an acidic greenhouse gas condensate of the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor.
 9. The thermal oxidization system of claim 8, wherein the greenhouse gas processor structurally configured to capture the liquid condensation of the at least one greenhouse gas from the oxidized gases within the greenhouse gas processor includes:
 - the greenhouse gas processor being structurally configured to acid neutralize the acidic greenhouse gas condensate by the greenhouse gas processor of the at least one greenhouse gas within the greenhouse gas processor.
 10. The thermal oxidization system of claim 1, wherein the oxidized gases received by the greenhouse gas processor includes at least one non-greenhouse gas; and
 - wherein the greenhouse gas processor is further structurally configured to vent the at least one non-greenhouse gas.
 11. The thermal oxidization system of claim 1, further comprising:
 - a thermocouple in one of:
 - thermal communication with the oxidation chamber to measure a temperature of the oxidation chamber, or
 - thermal communication with the retention chamber to measure a temperature of the retention chamber, or
 - thermal communication with the heat dissipator to measure a temperature of the heat dissipator.
 12. The thermal oxidization system of claim 1, further comprising at least one of:
 - a solenoid valve operable for regulating a feed of the waste gas stream into the oxidation mixer; and
 - an oxidant supply operable for regulating a feed of the oxidant into the oxidation mixer.
 13. The thermal oxidization system of claim 1, wherein the retention chamber includes:
 - a secondary heating element operable to facilitate a secondary combustion reaction of the combustible oxygenated waste gas stream into the oxidized gases within the retention chamber.
 14. The thermal oxidization system of claim 1, wherein the oxidation chamber further includes a supplemental air inlet structurally configured to facilitate a combustible mixture of an additional oxidant and the oxygenated waste gas stream flowing into the retention chamber.
 15. The thermal oxidization system of claim 1, further comprising:

25

an oxidation controller structurally configured to regulate an operation of the thermal oxidizer including at least one of:

- regulating the combustible mixture of the waste gas stream and the oxidant into the combustible waste gas stream within the oxidation mixer; 5
- regulating the primary combustion reaction of the combustible waste gas stream into an oxygenated waste gas stream within the oxidation chamber; and
- regulating the secondary combustion reaction of the oxygenated waste gas stream into the oxidized gases within the retention chamber. 10

16. The thermal oxidization system of claim **15**, further comprising:

- a thermocouple in thermal communication with the oxidation chamber to measure a temperature of the oxidation chamber; and 15

wherein, responsive to a measurement of the temperature of the oxidation chamber by the thermocouple, the oxidation controller is structurally configured to monitor the temperature of the oxidation chamber relative to at least one regulation threshold representative of a controlled operation of the thermal oxidizer. 20

17. The thermal oxidization system of claim **15**, further comprising: 25

- a thermocouple in thermal communication with the retention chamber to measure a temperature of the retention chamber; and

wherein, responsive to a measurement of the temperature of the retention chamber by the thermocouple, the oxidation controller is structurally configured to monitor the temperature of the retention chamber relative to at least one regulation threshold representative of a controlled operation of the thermal oxidizer. 30

26

18. The thermal oxidization system of claim **15**, further comprising:

- a thermocouple in thermal communication with the heat dissipater to measure a temperature of the heat dissipater; and

wherein, responsive to a measurement of the temperature of the heat dissipater by the thermocouple, the oxidation controller is structurally configured to monitor the temperature of the heat dissipater relative to at least one regulation threshold representative of a controlled operation of thermal oxidizer.

19. The thermal oxidization system of claim **15**, further comprising at least one of:

- a data logger structurally configured to log a regulation of the operation of the thermal oxidizer by the oxidation controller; and
- a data reporter structurally configured to remotely report the regulation of the operation of the thermal oxidizer by the oxidation controller.

20. The thermal oxidization system of claim **1**,

wherein the oxidation mixer is further structurally configured to facilitate an ignitable mixture of a natural gas stream and the oxidant within the oxidation mixer into an ignitable natural gas stream flowing with the oxidation mixer;

wherein a gas burner is in fluid communication with the oxidation mixer to receive a flow of the ignitable nature gas stream within the oxidation mixer; and

wherein the gas burner is structurally configured to ignite the ignitable nature gas stream within the oxidation chamber.

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